

SKYFIX

Predicting a high-altitude balloon's landing as a confidence ellipse, and re-estimating the flight's parameters from its own telemetry

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Use cases

ID	Use case	Actor	Command
UC-1	Pre-flight prediction: a footprint from catalogue values and a sounding	Recovery team lead	<code>predict --members 1000</code>
UC-2	In-flight re-prediction (primary): estimate the flight's parameters from telemetry and re-predict the footprint as it flies	Recovery team lead	<code>replay --log flight.csv</code>
UC-3	Post-flight scoring: how much did watching the flight help?	Payload engineer	<code>ReportService, T-V6</code>
UC-4	Model validation: the reference-case table as a CI gate	Examiner	<code>validate</code>
UC-5	Synthetic flight generation: twenty flights with known truth	Author	<code>synth</code>

1 • Introduction

A high-altitude balloon carries a payload to 30–45 km and drops it under a parachute, often two hundred kilometres downrange. A recovery team has to decide where to drive before the balloon bursts. They decide from a single pre-flight prediction computed with catalogue values for free lift, drag coefficient and burst diameter — each known only to within tens of per cent — and a wind profile that is already hours old at launch.

SKYFIX does two things about that. It predicts the landing as a **confidence ellipse** rather than a point, by flying a Monte Carlo ensemble over the dispersed parameters. And it **re-estimates the flight's real parameters from the telemetry as the flight happens**, with a particle filter, and re-predicts the footprint after each update.

Objectives and whether they were met

	Objective	Acceptance	Result
O1	Model the flight from published physics, not a curve fit	USSA-1976 within 0.1% of the reference table	met — worst 0.0107% over 25 altitudes
O2	Predict a footprint, not a point	50% and 95% ellipses from a dispersed ensemble	met — 1,000 members in 5.84 s against a 30 s budget
O3	Estimate flight parameters from telemetry	Burst altitude within 500 m on ≥ 18 of 20 flights	met — 20 of 20, seventeen inside 130 m
O4	Prove the in-flight update is worth it	Median landing-error reduction $\geq 30\%$	met — 74.0% at burst

Every number in this report was produced by a test in this repository and can be reproduced with the command named beside it. Nothing is quoted from memory.

2 • Problem statement

The full statement is in `statement.md`. The argument in short:

A point prediction misrepresents what is known. The inputs to a balloon flight prediction are known to tens of per cent, not to three figures. Free lift depends on how long someone held the fill valve open. Envelope drag depends on the envelope. Burst diameter is a catalogue figure for a population of latex balloons, not a property of the one on the launch rail. Propagating those through a nonlinear flight model and reporting a single latitude and longitude claims a precision that does not exist, and a recovery team that drives to it is being told something false.

The uncertainties that dominate are exactly the ones the flight reveals. A balloon climbing at 5.2 m/s rather than the predicted 4.8 is telling you about its lift and its drag. The altitude at which it bursts is a direct measurement of its burst diameter. That information arrives free, over a radio link the payload already carries, hours before it is needed — and a prediction that cannot take it in is throwing away the best data it will ever have.

The two halves interact. A parameter estimate with no uncertainty is as misleading as a point landing prediction, so the estimator has to report intervals, and those intervals have to be honest. §11 records what happened when they were not.

Target users

	Who	What they need
P1	Recovery team lead	Where to drive, and how large an area to plan for. Updated while the balloon is up.
P2	Payload engineer	The altitude profile against time — how long above 25 km, what the burst altitude was.
P3	Course examiner	Evidence that the physics, the persistence, the concurrency and the OOP are the author's own, and that the claims are tested.

Scope

In scope: a single balloon, a single sounding, replay of a recorded telemetry log, a command-line interface and exported charts. Out of scope, and each recorded as a decision rather than an oversight: 4-D gridded wind (ADR-7), terrain (constant ground elevation), live radio ingest (ADR-4), and a graphical interface (ADR-8).

3 · Functional requirements

Fourteen requirements in four milestones. Each states what goes in, what comes out, and what has to be true for it to count as done — the acceptance clause is what a test is written against, and several of them changed during implementation because a measurement contradicted them (FR-3.3 in ADR-17, T-V3's tolerance in ADR-13).

M1 — Ingest and configuration

FR-1.1 Sounding ingest. In: a University-of-Wyoming-format text profile. Out: station, epoch and N `SoundingLevel` rows in pressure order. Accept: a 1,200-level file in ≤ 2 s; every rejected line reported with `file:line:reason` and never silently dropped.

FR-1.2 Telemetry ingest, replay only. In: a CSV log with epoch, packet id, position, altitude, pressure. Out: an ordered, de-duplicated `TelemetrySeries` with quality flags. Accept: duplicate packet ids, out-of-order timestamps and GPS dropouts are counted and reported; a dropout with a pressure reading gets a pressure altitude, flagged as such.

FR-1.3 Configuration validated by rule. In: mission and balloon JSON. Out: an immutable `BalloonConfig`. Accept: nine stated rules, each producing a `ValidationException` naming the offending field and value; exit code 2; no stack trace shown.

FR-1.4 Synthetic flight generation. In: a seed and a parameter draw. Out: a noisy telemetry log plus the truth it was generated from. Accept: the same seed reproduces a byte-identical file, and the stored truth is what T-V5 and T-V6 score against.

M2 — Forward model

FR-2.1 Atmosphere. In: geometric altitude. Out: temperature, pressure, density. Accept: within 0.1% of the USSA-1976 reference table at 25 altitudes (T-V1); a query above 86 km is refused with a `ModelDomainException` rather than

extrapolated.

FR-2.2 Wind field. In: a sounding and an altitude. Out: an east-north wind vector. Accept: linear interpolation in height; a query outside the profile's range is held at the nearest value **and flagged**, and the flight record counts how many steps used a held value.

FR-2.3 Flight model. In: configuration, parameters, launch point. Out: a state history from launch to landing. Accept: ascent rate within 2% of the analytic buoyancy–drag terminal velocity (T-V2); RK4 and RKF45 agree on the landing point (T-V3); gas mass conserved (T-V4).

FR-2.4 Dispersed ensemble and fitted ellipse. In: a dispersion specification and a member count. Out: every member's landing point and 50% / 95% confidence ellipses. Accept: 1,000 members in ≤ 30 s on four cores (T-P1); the same seed reproduces identical semi-axes and orientation to $1e-9$ (T-R1); empirical containment inside 0.90–0.98 (T-V7).

M3 — In-flight estimator

FR-3.1 Measurement model. In: a predicted state and a telemetry sample. Out: a Gaussian log-likelihood over altitude, vertical rate and horizontal position with a configurable sigma per channel. Accept: maximised at the injected truth, monotone decreasing with induced error in each channel.

FR-3.2 Particle filter update. In: a particle set over $\theta =$ (free lift, ascent Cd, burst-diameter scale, parachute Cd) and telemetry to time t. Out: a posterior with per-parameter bands, plus per-update statistics persisted. Accept: T-V5's thresholds; ESS, resample count and jitter logged at each update.

FR-3.3 Re-prediction on update. In: a posterior and the current measured state. Out: a forward ensemble from that state, tagged with the update epoch. Accept: one 200-member re-prediction in ≤ 5 s, issued no more often than once per 50 s of flight time, so a 1 Hz log replayed at $10\times$ drops no updates (T-P2). **The interval is derived from the budget rather than chosen** — see ADR-17.

FR-3.4 Burst detection. In: the replay stream. Out: a burst event, and the moment burst diameter becomes measurable. Accept: detected within 5 s and 150 m under 10 m GPS noise, with zero false positives across a 3 s dropout (T-E2).

M4 — Comparison and reporting

FR-4.1 Prediction error scoring. In: a frozen run, a replay run, and the actual landing point. Out: haversine error per update and the median reduction for O4. Accept: matches a hand-computed haversine within 1 m.

FR-4.2 Plot export. In: a run id. Out: PL-1...PL-6 as PNGs in `out/run-<id>/`.

Implementation and verification map

ID	Requirement	Implemented by	Verified by
FR-1.1	Sounding ingest with per-line rejection	WyomingSoundingReader , SoundingDao	T-U-WIND, T-E3
FR-1.2	Telemetry ingest, replay-only	CsvTelemetryReader , TelemetryDao	T-E5, T-P4
FR-1.3	Configuration validated by rule	BalloonConfig.Builder , ConfigLoader	T-U-BUILDER, T-E1
FR-1.4	Synthetic flight generation with stored truth	SyntheticFlightWriter , SynthService	T-R1, T-V5
FR-2.1	USSA-1976 atmosphere, 0–86 km	Ussa1976Atmosphere	T-V1
FR-2.2	Wind interpolation from one sounding	SoundingWindField	T-U-WIND
FR-2.3	Ascent / burst / descent flight model	FlightPhase and subclasses, FlightSimulator	T-V2, T-V3, T-V4
FR-2.4	Dispersed ensemble and fitted ellipse	EnsembleRunner , EllipseFitter	T-P1, T-R1, T-V7
FR-3.1	Measurement model	GaussianMeasurementModel	T-U-LIKELIHOOD
FR-3.2	Particle filter update	ParticleFilter , FilterBank, EstimateDao	T-V5
FR-3.3	Re-prediction on update	ReplayService , EnsembleRunner	T-P2, T-V6
FR-3.4	Burst detection	BurstDetector	T-E2
FR-4.1	Prediction error scoring	ReportService	T-V6
FR-4.2	Plot export PL-1...PL-6	PlotExporter	PlotExporterTest

4 • Non-functional requirements

NFR-1 Performance. A footprint has to be computable while a team is still at the launch site, and a re-prediction has to keep up with a replay running faster than real time.

NFR-2 Reproducibility. A result that cannot be reproduced is not evidence. The same seed must produce identical ellipse parameters regardless of how many threads ran, which is a real constraint on the concurrency design: it rules out a shared generator entirely.

NFR-3 Error handling. Every external input failure produces a typed exception carrying `file:line:field`, a documented non-zero exit code, and no stack trace shown to the user. A ten-thousand-line corrupted sounding and a truncated binary dump both have to exit cleanly.

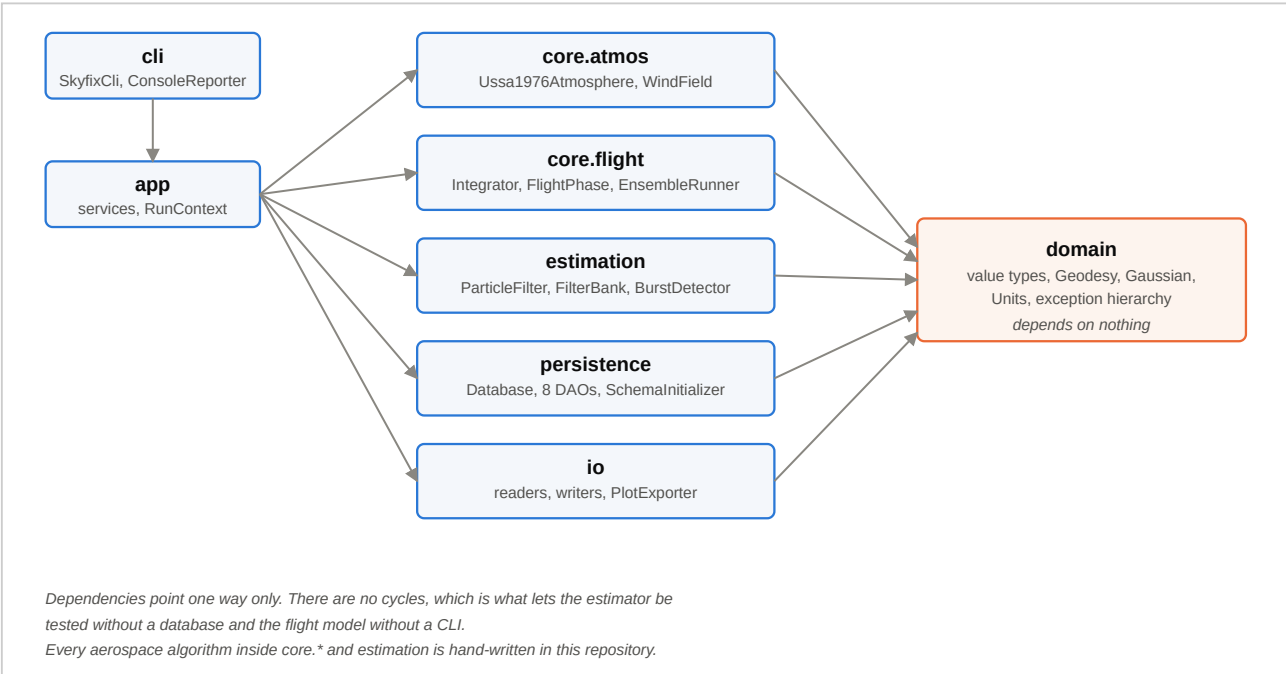
NFR-4 Offline operation. A fresh clone plus a JDK runs every command and every test with no network access. All data ships in `data/`. A test that needs the network is a bug.

NFR-5 Coverage. 80% line coverage on the packages that carry the physics and the estimator. The CLI, io and persistence packages are exempt from the strict gate and covered instead at the command surface by T-U1, T-E1 and T-E3.

ID	Requirement	Budget	Measured
NFR-1	1,000-member ensemble on 4 cores	30 s	5.84 s (3-run median, warm JVM, sounding wind)
NFR-1	200-member re-prediction	5 s	0.18 s mean over 140 re-predictions
NFR-2	Reproducibility: same seed, same ellipse	1e-9	met , 1 thread against 4
NFR-3	Typed errors carrying <code>file:line:field</code> , no stack traces	—	T-E1, T-E3, six corrupt-input cases
NFR-4	Offline: no command or test touches the network	—	enforced by <code>skyfix.offline</code> and T-S1
NFR-5	Coverage on <code>core.*</code> and <code>estimation</code>	80% line	100% / 93.6% / 91.7%

5 • System architecture

Strict one-way dependencies: `cli` → `app` → {`core.*`, `estimation`, `persistence`, `io`} → `domain`, and `domain` depends on nothing. There are no cycles, which is what lets the estimator be tested without a database and the flight model without a CLI.



Package dependencies. One direction only; domain depends on nothing.

Every aerospace algorithm is hand-written in this repository: the atmosphere model, RK4 and RK45, the flight dynamics, sounding interpolation, Latin-hypercube sampling, the covariance eigen-decomposition, the error function, the particle filter, systematic resampling and the Cholesky factor the jitter kernel uses. Third-party code is confined to plumbing — SQLite JDBC, Jackson for configuration JSON, XChart for PNG export, JUnit and AssertJ for tests.

6 • Design diagrams

`docs/diagrams/` holds the architecture diagram above and the replay sequence below. The sequence was **corrected during week 9** — see ADR-17 — because the version in BLUEPRINT §8 drew the re-prediction inside the per-sample loop, which the requirement's own timing budget makes impossible. CLAUDE.md's sixth rule is that a stale diagram is worse than no diagram, so the correction landed in the same commit as the code.

```

%% SKYFIX replay loop (FR-3.2, FR-3.3, FR-3.4).
%% Corrected per ADR-17: the re-prediction sits in a guarded block, not in the per-sample
loop.
%% A 10x replay of a 1 Hz log affords 100 ms per sample; a 5 s re-prediction cannot run per
sample.
sequenceDiagram
    actor User
    participant CLI as SkyfixCli
    participant RS as ReplayService
    participant TR as CsvTelemetryReader
    participant BD as BurstDetector
    participant FB as FilterBank
    participant ER as EnsembleRunner
    participant DB as DAOs

    User->>CLI: run.sh replay --log flight.csv --every 20
    CLI->>RS: replay(mission, config, series, options, context)
    RS->>DB: save(RunRecord kind=REPLAY, seed, gitSha, configHash)
    DB-->>RS: runId
    RS->>TR: read(logPath)
    TR-->>RS: TelemetrySeries
    RS->>FB: start(launch, launchEpoch)

    loop per assimilated sample (FR-1.2)
        RS->>BD: observe(sample)
        alt burst detected (FR-3.4)
            BD-->>RS: BurstEvent(epoch, altitude)
            RS->>FB: burstObserved()
        end
        RS->>FB: update(observation)
        par each independent filter (ADR-18)
            FB->>FB: propagate, weight (FR-3.1), ESS check
            FB->>FB: systematic resample + Liu-West jitter if ESS < N/2
        end
        FB-->>RS: pooled Posterior(median, p05, p95, ESS)

        alt 50 s of flight time elapsed, or burst, or last airborne sample (ADR-17)
            RS->>ER: runFrom(measured state, posterior spec, 200 members)
            ER-->>RS: LandingEllipse(50%, 95%)
            RS->>DB: saveEllipse(runId, epoch, ellipse)
        end
    end

    RS->>DB: saveAll(runId, posteriors)
    RS->>DB: finish(runId, wallClockMs, OK)
    RS-->>CLI: ReplayResult
    CLI-->>User: posterior bands, burst, footprint, PL-3

```

The two loops the diagram separates have very different costs. A filter update is milliseconds — at 500 particles it costs about 2 ms per second of flight time, against a 100 ms budget at 10× replay. A 200-member re-prediction is seconds. Running both per sample is what the original diagram implied and what the arithmetic forbids.

7 • Design decisions and rationale

Eighteen decision records live in `docs/adr/`. The five that most shaped the result:

ADR-1 — USSA-1976 rather than an exponential atmosphere. Measured: a best-fit single-scale-height exponential is 20.4% off in density at the tropopause and 48.3% off by 35 km. That error would have swamped everything downstream.

ADR-13 — the integration step is bounded by descent stability, not by accuracy. Quadratic drag linearises to a real eigenvalue whose magnitude grows as the payload falls into denser air, reaching about 3.6 /s near the ground. RK4's real-axis stability limit is 2.785, capping the step at ~0.77 s there. At the blueprint's original 1 s step the landing came out 39 s late and about 470 m off, with the descent rate visibly oscillating. The default is now 0.25 s and the simulator *refuses* an unstable step rather than returning a plausible-looking wrong answer. Each integrator derives its own limit by applying itself to $y' = \lambda y$; nothing is transcribed.

ADR-3 — a bootstrap particle filter, and what it cannot recover. Burst is a hard discontinuity and the pre-burst burst-diameter posterior is genuinely uninformative, which no Kalman variant represents honestly. The record also carries the identifiability analysis that limits the result — see §11.

ADR-17 — re-prediction runs on a flight-time interval, not per sample. FR-3.3 allowed a re-prediction 5 s and justified it by a 10× replay dropping no updates. A 10× replay of a 1 Hz log delivers a sample every 100 ms, so a per-sample 5 s re-prediction is fifty times over budget — the criterion was unsatisfiable as drawn. It closes under one reading: 100 ms per second of flight buys one 5 s re-prediction per 50 s of flight. The two loops are now separate.

ADR-18 — the posterior is pooled from a bank of independent filters. The decision this project turned on; see §11.

All eighteen decisions

ID	Decision	Chosen over	Why
ADR-1	USSA-1976 atmosphere	exponential; NRLMSISE-00	Offline and closed-form, and the published table becomes the T-V1 oracle. An exponential is 48.3% off in density by 35 km.
ADR-2	SQLite with hand-written DAOs	H2; MySQL; JPA/Hibernate	The examiner must see the author's JDBC and SQL; one file, zero install.
ADR-3	Bootstrap particle filter	EKF; UKF; batch least squares	Burst is a hard discontinuity and the pre-burst posterior is genuinely uninformative; no Jacobians needed.
ADR-4	Replay-only telemetry	live serial; UDP listener	An examiner with no hardware must still exercise the estimator, and replay makes it deterministic.
ADR-5	Explicit <code>ExecutorService</code> , immutable state	parallel streams; single-threaded	The syllabus requires concurrency and the examiner should see pool lifecycle, futures and shutdown.
ADR-6	One derived seed per member	shared synchronised <code>Random</code>	A shared generator makes results depend on thread interleaving and breaks T-R1.
ADR-7	Single-sounding wind field	GFS GRIB2; ERA5	GRIB2 plus a mandatory download breaks the offline rule; the error is named and carried by the <code>wind_scale</code> dispersion.
ADR-8	CLI plus exported PNGs	JavaFX; Swing	No GUI in the lab record; captures come from the charts and console tables.
ADR-9	Spherical advection, haversine scoring	WGS-84 geodesic; flat earth	Measured: 0.327% against a Vincenty inverse solution at mission scale, far below the wind error that dominates.
ADR-10	Config identity by SHA-256	config id alone	A run is reproducible only if the exact configuration is pinned.
ADR-11	Free lift drives inflation	launch diameter drives it	Free lift is what a team actually sets at the fill valve.
ADR-12	RKF45 as a fixed-step cross-check	adaptive stepping	T-V3 compares two schemes; adaptive stepping would compare two different discretisations as well.
ADR-13	Step bounded by descent stability	a smaller step by habit	Measured: RK4 at 1 s leaves its stability region below ~6 km and lands 39 s late.
ADR-14	Covariance ellipse, aspect ratio reported	convex hull; density contour	Simple, closed-form and testable; the aspect ratio exposes where the assumption strains.
ADR-15	Timing measured without instrumentation	one <code>verify</code> run for both	A figure measured under a bytecode agent is not a performance figure.
ADR-16	One chart palette and chrome	per-chart styling	The plots read as one set; colours validated once against the print surface.
ADR-17	Re-prediction on a flight-time interval	per telemetry sample	Measured: a per-sample 5 s re-prediction is fifty times over FR-3.3's own budget.
ADR-18	Posterior pooled from a filter bank	one filter with more particles	Measured: one filter's band covered the truth 0/20 times; eight seeds disagreed by 100× their own band width.

8 · Implementation details

Datasets

ID	What	Provenance
DS-1	Radiosonde sounding	<code>data/soundings/</code> ships a synthetic profile, labelled as such in its header, because the upper-air archive was not reachable from the build environment. <code>[PLACEHOLDER – a real sounding, once one can be downloaded and its licence checked.]</code>
DS-3	USSA-1976 reference table, 25 altitudes	Generated from two independent third-party implementations of the standard that agree to 0.00987% . Carries a <code>verify:</code> marker requiring hand transcription from NOAA-S/T 76-1562 Table I.
DS-4	Balloon catalogue figures	<code>data/missions/balloon.json</code> carries a <code>verify:</code> marker on the burst and launch diameters.
DS-6	Twenty synthetic flights with known truth	<code>data/truth/seeds.csv</code> is the definition; the telemetry regenerates from it byte-identically and is not committed. Every generated file says in its first line that it is synthetic.

Concurrency

Two explicit fixed pools, both with a `CompletionService` and a deliberate `shutdown()` / `awaitTermination()`: `EnsembleRunner` flies ensemble members, `FilterBank` advances independent filters. There is no shared mutable `Random` anywhere — member k derives its own seed from the run seed, so member k is reproducible in isolation and the result does not depend on thread interleaving. T-R1 checks that one thread and four threads produce identical ellipse parameters to $1e-9$.

Persistence

Thirteen tables, plain JDBC, the DAO pattern, `PreparedStatement` everywhere. Physics is enforced in the schema as CHECK constraints — a parachute Cd outside $[0.1, 2.0]$ or an ellipse with a non-positive semi-major axis cannot be stored, so a bug that produces one fails at the database rather than being persisted and reported. Test **T-S1** scans `src/main` for SQL built by string concatenation and fails the build if it finds any.

Table	Holds	Notable constraint
<code>schema_version</code>	applied migrations	—
<code>mission</code>	launch site, ground elevation, epoch	latitude and longitude in range
<code>balloon_config</code>	the configuration and its SHA-256	drag coefficients in $[0.1, 2.0]$; masses positive
<code>sounding</code>	station, epoch, source, file hash	unique on (station, epoch, source)
<code>sounding_level</code>	pressure, height, wind per level	pressure descending within a sounding
<code>flight_log</code>	an ingested telemetry log and its hash	unique on (mission, name)
<code>telemetry_sample</code>	one row per packet, with quality flags	unique packet id within a log
<code>run</code>	seed, git SHA, config hash, integrator, step, member count, host cores, wall clock	<code>run_kind</code> and <code>integrator</code> constrained to known values
<code>ensemble_member</code>	every member's parameters and landing point	—
<code>run_state</code>	sampled trajectory states	phase constrained
<code>estimate</code>	one row per parameter per update	<code>p05 <= median <= p95</code> ; parameter name constrained
<code>landing_ellipse</code>	fitted footprints, optionally epoch-tagged	semi-axes positive; azimuth in $[0, 360]$
<code>validation_result</code>	the reference-case table, per run	—

Every run stores its seed, git SHA, configuration hash, integrator, step size, member count, host core count and wall-clock time. That is what makes SC-6 possible: the same seed reproduces the same ellipse, and the row proves which code and

which configuration produced it.

Class inventory

One hundred classes in `src/main`, by package:

Package	Principal types
domain	GeoPoint, BalloonState, StateHistory, BalloonConfig + Builder, FlightParameters, SimSettings, DispersionSpec, Distribution, Ensemble, LandingEllipse, Posterior, PredictionError, TelemetrySample, TelemetrySeries, FlightTruth, NoiseSpec, Geodesy, Gaussian, Units, LiftGas, Phase
domain.error	SkyfixException and five subclasses, each mapping to an exit code
core.atmos	AtmosphereModel, Ussa1976Atmosphere, ExponentialAtmosphere, WindField, ConstantWindField, SoundingWindField, WindFieldFactory
core.flight	Integrator, Rk4Integrator, Rkf45Integrator, IntegratorFactory, FlightPhase, AscentPhase, DescentPhase, FlightSimulator, DispersionSampler, EnsembleRunner, EllipseFitter
estimation	MeasurementModel, GaussianMeasurementModel, Observation, Particle, ParticleFilter, FilterBank, Resampler, SystematicResampler, WeightedQuantile, BurstDetector, BurstEvent
persistence	Database, SchemaInitializer, Repository, and eight DAOs
io	SoundingReader, WyomingSoundingReader, CsvTelemetryReader, CsvWriter, GeoJsonWriter, ConfigLoader, SyntheticFlightWriter, PlotExporter, ChartStyle, FileDigest
app	IngestService, PredictionService, ReplayService, ReportService, SynthService, ValidationService, ReplayOptions, RunContext
cli	Main, SkyfixCli, ConsoleReporter

The algorithms, and why each is hand-written

CLAUDE.md's first rule for this project is that every aerospace algorithm is the author's own Java. No Orekit, no Apache Commons Math, no scientific library anywhere in `src/main`. What that means in practice:

USSA-1976 atmosphere. Only the seven layers' base temperatures and lapse rates are transcribed; every base pressure is computed by recursion from sea level, because transcribing eight pressures is eight chances to introduce a typo that no test would catch. Within a layer the barometric formula is applied in geopotential height, $H = r h / (r + h)$ with the standard effective earth radius. Measured against the reference table: worst 0.0107% in density over 25 altitudes.

RK4 and RKF45. Both written out from their tableaux. Each derives its own real-axis stability limit numerically, by applying itself to $y' = \lambda y$ at unit step and bisecting for $|R| = 1$ — RK4 comes out at 2.785293563, RKF45 at 3.678. Nothing is transcribed, so a mistyped constant cannot hide, and ADR-13's stability guard is built on a number the code computed about itself.

Latin hypercube sampling. N strata per dimension, one sample in each, an independent permutation per dimension. The dimension order is part of the reproducibility contract: reordering it would change every member's parameters for the same seed and break T-R1.

The confidence ellipse. A closed-form 2×2 symmetric eigen-decomposition of the landing covariance, scaled by the chi-square factor for a bivariate normal, $s = \sqrt{-2 \ln(1 - p)}$, which gives 1.1774 at 50% and 2.4477 at 95%. §11 is about what happens when the cloud is not bivariate normal.

The error function. `Gaussian.erf` switches at $|x| = 2$ from a Maclaurin series to a continued fraction evaluated by modified Lentz, because the series loses to cancellation past $|x| \approx 3$. The first implementation used the series alone and `cdf(8)` was wrong in the fifth decimal — invisible, because the test only swept ± 4 . The test now sweeps ± 8 and checks the two methods agree at the crossover.

The particle filter. Propagation, log-space weighting, Kish's effective sample size, systematic resampling by a single uniform and a regular comb, and Liu–West shrinkage jitter whose kernel is the full weighted covariance factorised by a hand-written Cholesky decomposition.

Geodesy. Haversine distance and bearing, with the spherical assumption's error *measured* against a Vincenty inverse solution on the WGS-84 ellipsoid rather than assumed small: 0.327% at mission scale (ADR-9), re-measured on every build so it cannot drift.

Object-oriented design the examiner should look at

Template Method — `FlightPhase` fixes the sequence every phase follows (query the atmosphere, query the wind, form the net vertical force, advect horizontally) and subclasses supply only what differs between ascending under a balloon and descending under a parachute. Burst becomes a change of phase object rather than a branch threaded through the simulator.

Strategy — `Integrator`, `MeasurementModel`, `Resampler`, `WindField` and `AtmosphereModel` are all interfaces with more than one real implementation, and in each case the alternative exists to be *compared against* in a test rather than to satisfy a pattern checklist.

Builder with validation — `BalloonConfig.Builder` and `SimSettings.Builder` validate in `build()` and throw a `ValidationException` naming the failing field, so an invalid object cannot be constructed at all.

Immutable value types — every domain type is a record or a final class. That is what makes the ensemble safe to run across threads without a single `synchronized` block.

9 · Screenshots and results

All of the following were produced after `v0.4-validated` by the commands shown. The console blocks are verbatim stdout, included from `docs/screenshots/` at build time so the report cannot disagree with the run that produced it. The charts are the exporters' own PNGs, 1460 px or wider.

SC-1 — the reference-case validation table

```
$ skyfix validate
```

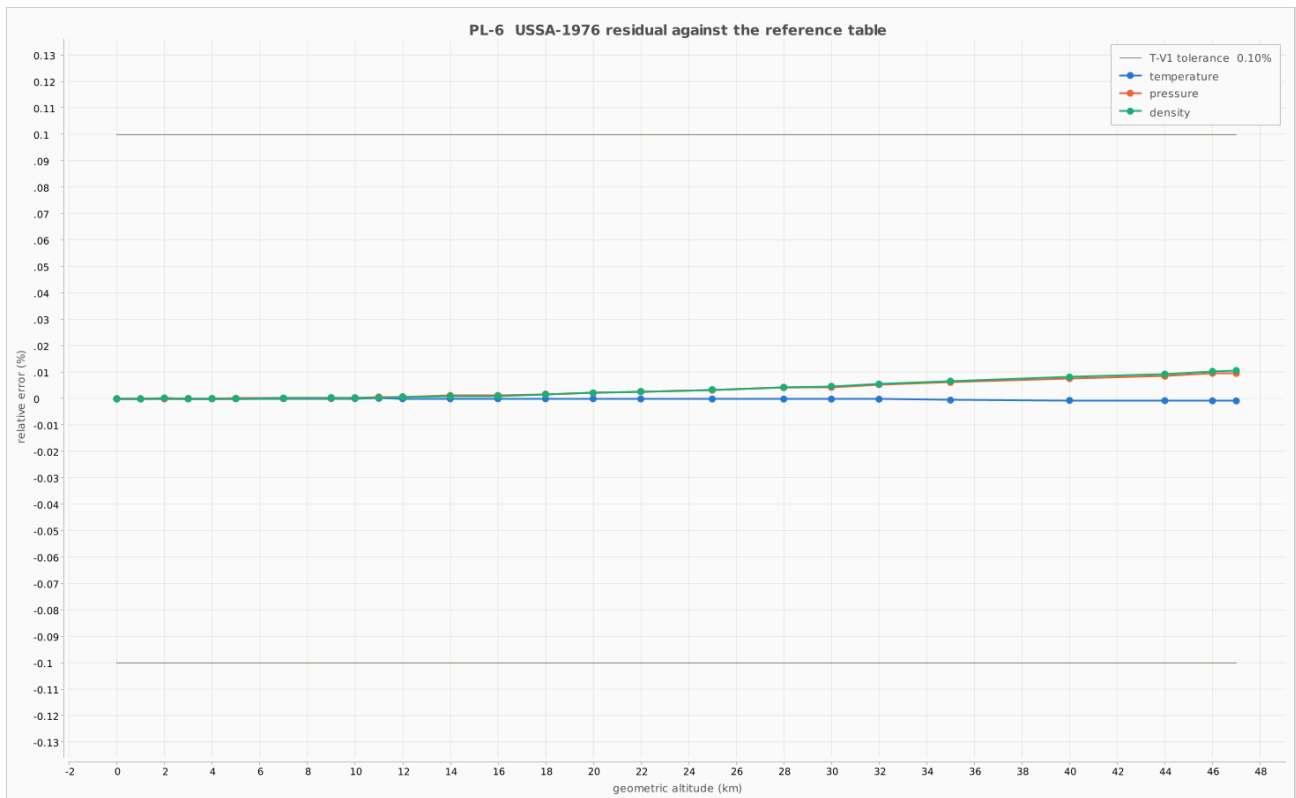
```
SKYFIX reference-case validation
```

CASE	QUANTITY	MODEL	REFERENCE	ERROR	RESULT
T-V1	temperature @ 0 m	288.150	288.150	0.00000%	PASS
T-V1	pressure @ 0 m	101325	101325	0.00000%	PASS
T-V1	density @ 0 m	1.22500	1.22500	0.0001%	PASS
T-V1	temperature @ 1000 m	281.651	281.651	0.00000%	PASS
T-V1	pressure @ 1000 m	89876.3	89876.3	0.00000%	PASS
T-V1	density @ 1000 m	1.11166	1.11166	0.0001%	PASS
T-V1	temperature @ 2000 m	275.154	275.154	0.00000%	PASS
T-V1	pressure @ 2000 m	79501.4	79501.4	0.00000%	PASS

T-V1	density @ 2000 m	1.00655	1.00655	0.0003%	PASS
T-V1	temperature @ 3000 m	268.659	268.659	0.0000%	PASS
T-V1	pressure @ 3000 m	70121.2	70121.1	0.0001%	PASS
T-V1	density @ 3000 m	0.909254	0.909254	0.0000%	PASS
T-V1	temperature @ 4000 m	262.166	262.166	0.0000%	PASS
T-V1	pressure @ 4000 m	61660.4	61660.4	0.0001%	PASS
T-V1	density @ 4000 m	0.819346	0.819346	0.0000%	PASS
T-V1	temperature @ 5000 m	255.676	255.676	0.0000%	PASS
T-V1	pressure @ 5000 m	54048.3	54048.2	0.0002%	PASS
T-V1	density @ 5000 m	0.736428	0.736428	0.0001%	PASS
T-V1	temperature @ 7000 m	242.700	242.700	0.0001%	PASS
T-V1	pressure @ 7000 m	41105.3	41105.2	0.0002%	PASS
T-V1	density @ 7000 m	0.590018	0.590017	0.0002%	PASS
T-V1	temperature @ 9000 m	229.733	229.733	0.0001%	PASS
T-V1	pressure @ 9000 m	30800.7	30800.6	0.0003%	PASS
T-V1	density @ 9000 m	0.467063	0.467061	0.0004%	PASS
T-V1	temperature @ 10000 m	223.252	223.252	0.0001%	PASS
T-V1	pressure @ 10000 m	26499.9	26499.8	0.0004%	PASS
T-V1	density @ 10000 m	0.413510	0.413509	0.0003%	PASS
T-V1	temperature @ 11000 m	216.774	216.773	0.0001%	PASS
T-V1	pressure @ 11000 m	22700.0	22699.8	0.0007%	PASS
T-V1	density @ 11000 m	0.364802	0.364800	0.0004%	PASS
T-V1	temperature @ 12000 m	216.650	216.650	0.0000%	PASS
T-V1	pressure @ 12000 m	19399.4	19399.3	0.0008%	PASS
T-V1	density @ 12000 m	0.311938	0.311936	0.0007%	PASS
T-V1	temperature @ 14000 m	216.650	216.650	0.0000%	PASS
T-V1	pressure @ 14000 m	14170.4	14170.2	0.0013%	PASS
T-V1	density @ 14000 m	0.227856	0.227854	0.0009%	PASS
T-V1	temperature @ 16000 m	216.650	216.650	0.0000%	PASS
T-V1	pressure @ 16000 m	10352.8	10352.7	0.0013%	PASS
T-V1	density @ 16000 m	0.166471	0.166469	0.0011%	PASS
T-V1	temperature @ 18000 m	216.650	216.650	0.0000%	PASS
T-V1	pressure @ 18000 m	7565.23	7565.10	0.0018%	PASS
T-V1	density @ 18000 m	0.121647	0.121645	0.0017%	PASS
T-V1	temperature @ 20000 m	216.650	216.650	0.0000%	PASS
T-V1	pressure @ 20000 m	5529.31	5529.19	0.0022%	PASS
T-V1	density @ 20000 m	0.0889099	0.0889080	0.0022%	PASS
T-V1	temperature @ 22000 m	218.574	218.574	0.0001%	PASS
T-V1	pressure @ 22000 m	4047.50	4047.39	0.0027%	PASS
T-V1	density @ 22000 m	0.0645098	0.0645081	0.0027%	PASS
T-V1	temperature @ 25000 m	221.552	221.552	0.0001%	PASS
T-V1	pressure @ 25000 m	2549.22	2549.14	0.0033%	PASS
T-V1	density @ 25000 m	0.0400839	0.0400825	0.0035%	PASS
T-V1	temperature @ 28000 m	224.527	224.528	0.0001%	PASS
T-V1	pressure @ 28000 m	1616.20	1616.13	0.0042%	PASS
T-V1	density @ 28000 m	0.0250763	0.0250752	0.0044%	PASS
T-V1	temperature @ 30000 m	226.509	226.509	0.0001%	PASS
T-V1	pressure @ 30000 m	1197.03	1196.98	0.0043%	PASS
T-V1	density @ 30000 m	0.0184102	0.0184093	0.0047%	PASS
T-V1	temperature @ 32000 m	228.490	228.490	0.0002%	PASS
T-V1	pressure @ 32000 m	889.064	889.017	0.0053%	PASS
T-V1	density @ 32000 m	0.0135552	0.0135544	0.0055%	PASS
T-V1	temperature @ 35000 m	236.513	236.515	0.0005%	PASS
T-V1	pressure @ 35000 m	574.595	574.559	0.0062%	PASS
T-V1	density @ 35000 m	0.00846338	0.00846281	0.0067%	PASS
T-V1	temperature @ 40000 m	250.350	250.351	0.0006%	PASS

T-V1	pressure @ 40000 m	287.144	287.122	0.0076%	PASS
T-V1	density @ 40000 m	0.00399568	0.00399535	0.0082%	PASS
T-V1	temperature @ 44000 m	261.403	261.405	0.0007%	PASS
T-V1	pressure @ 44000 m	169.496	169.482	0.0085%	PASS
T-V1	density @ 44000 m	0.00225885	0.00225864	0.0093%	PASS
T-V1	temperature @ 46000 m	266.925	266.927	0.0008%	PASS
T-V1	pressure @ 46000 m	131.341	131.328	0.0096%	PASS
T-V1	density @ 46000 m	0.00171415	0.00171397	0.0104%	PASS
T-V1	temperature @ 47000 m	269.684	269.686	0.0008%	PASS
T-V1	pressure @ 47000 m	115.851	115.840	0.0096%	PASS
T-V1	density @ 47000 m	0.00149652	0.00149636	0.0107%	PASS
T-V2	ascent rate @ 2000 m	3.94258	3.94272	0.0036%	PASS
T-V2	ascent rate @ 8001 m	4.39323	4.39346	0.0052%	PASS
T-V2	ascent rate @ 14001 m	5.05002	5.05048	0.0092%	PASS
T-V2	ascent rate @ 20000 m	5.90730	5.90804	0.0125%	PASS
T-V2	ascent rate @ 26002 m	6.92503	6.92622	0.0172%	PASS
T-V3	RK4 vs RKF45 landing @ dt=0.500 s	0.00254561	0.000000	0.002546	PASS
T-V3	RK4 vs RKF45 landing @ dt=0.250 s	8.53528e-05	0.000000	8.535e-05	PASS
T-V3	RK4 vs RKF45 landing @ dt=0.125 s	4.84572e-06	0.000000	4.846e-06	PASS
T-V4	gas-law mass invariant, worst rel...	7.91290e-16	0.000000	7.913e-16	PASS

T-V1	75 checks, tolerance 0.100%	worst 0.01071%	PASS		
T-V2	5 checks, tolerance 2.000%	worst 0.01719%	PASS		
T-V3	3 checks, tolerance 50.00 m	worst 0.002546	PASS		
T-V4	1 checks, tolerance 1.000e-06	1 worst 7.913e-16	PASS		
Not yet implemented (reported, not silently passed):					
- T-V5 parameter recovery over 20 synthetic flights - needs the particle filter (FR-3.2), due W8					
- T-V6 live-vs-frozen landing error reduction - needs the estimator (FR-3.3), due W9					
- T-V7 95% ellipse empirical containment - needs the ensemble (FR-2.4), due W5					
84 checks, 84 passed, 0 failed					



PL-6. The atmosphere model's residual against the DS-3 table, by altitude, against the T-V1 tolerance band. T-V1 reports one worst-case number; this shows whether the error is uniform or merely small on average.

SC-2 — ingest, naming what it rejected

```
$ skyfix ingest --sounding data/soundings/SYNTHETIC_2026-09-14_00Z.txt --epoch 2026-09-14T00:00:00Z --mission data/missions/mission.json --balloon data/missions/balloon.json --db skyfix-report.db
sounding SYNTH @ 2026-09-14T00:00:00Z: 35 levels, 0 lines rejected
sounding id: 1
mission "HabSat-1" id=1 (new), config b9997682e191 id=1 (new)
```

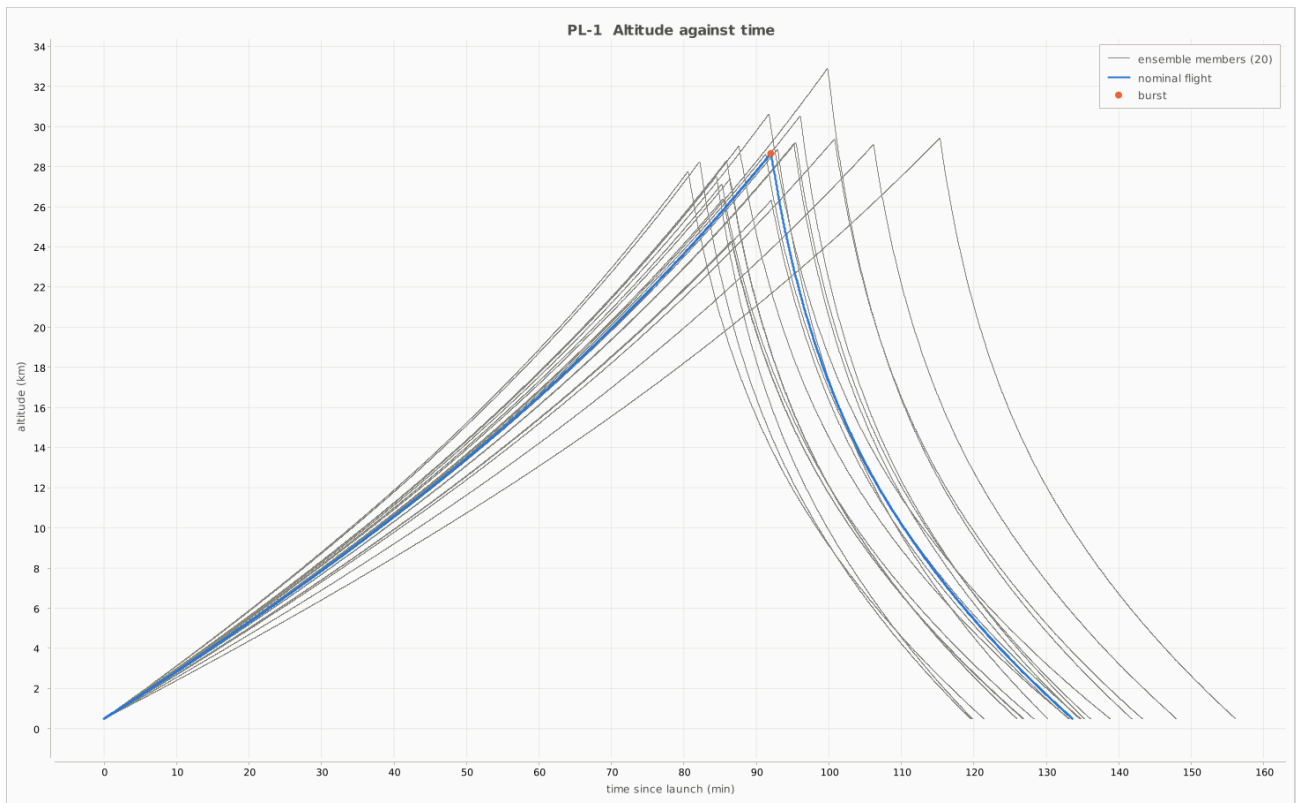
SC-3 — a 1,000-member footprint

```
$ skyfix predict --members 1000 --sounding-id 1

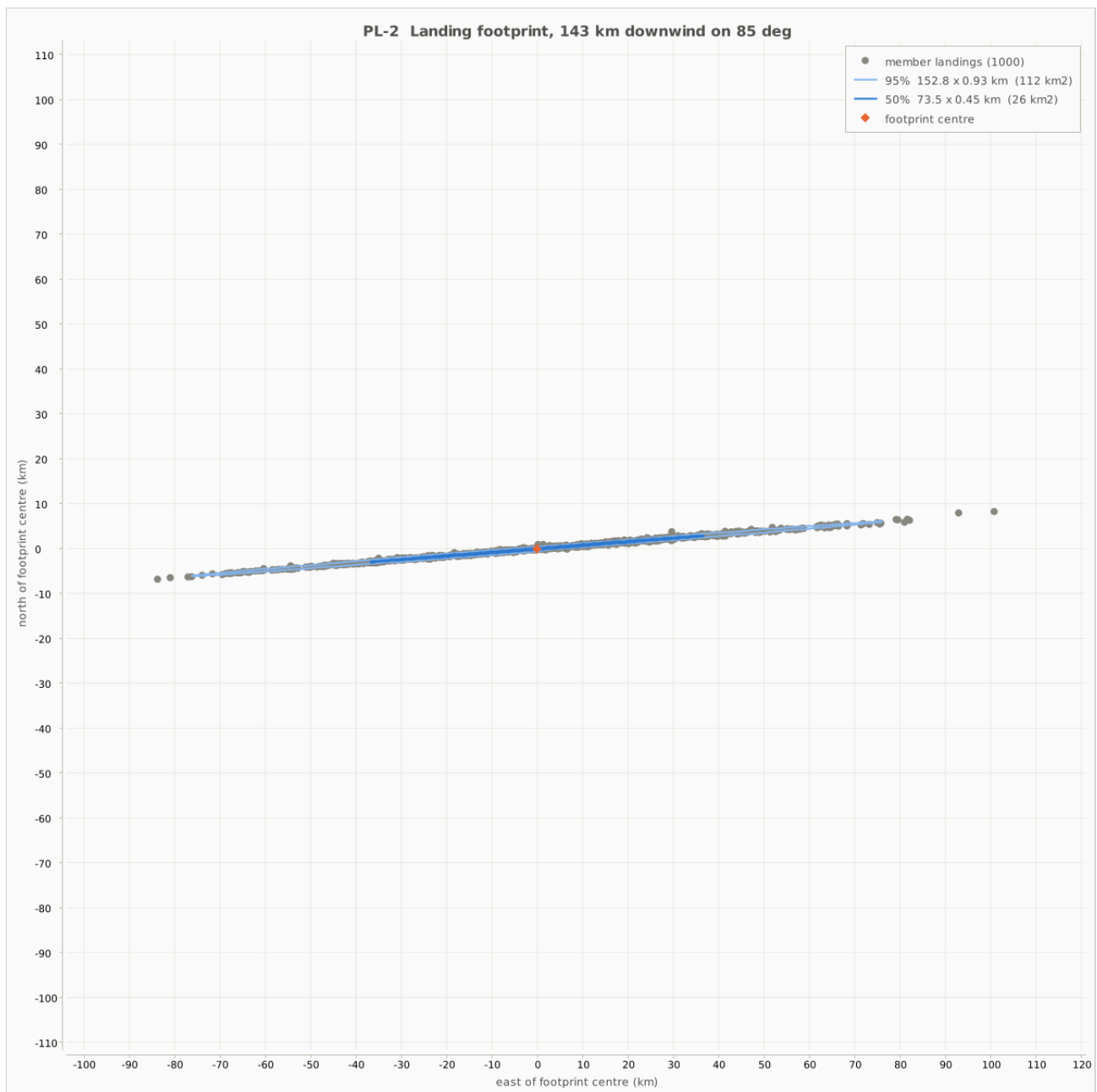
Run 1 - landing footprint, 1000 members (wind: SOUNDING)
-----
nominal burst      28,668 m    at T+5522 s
mean burst         28,586 m
flight time        8,021 s    (134 min)

50% ellipse  centre  23.363580, 78.812592
                axes  36.7 km x 224 m at 85 deg | area 25.81 km2
95% ellipse  centre  23.363580, 78.812592
                axes  76.4 km x 465 m at 85 deg | area 111.57 km2

ensemble          5,208 ms on 4 threads
exports           out/run-1
seed 20260914 | git 0b4960585bcd6ed72d1cd17ca43beb279a00aa8a | RK4 dt=0.25 s | 1000 members
| 6540 ms
```



PL-1. Altitude against time, with the retained ensemble members behind the nominal flight. The fan opens on the ascent as members diverge in burst altitude, then the descents run roughly parallel.



PL-2. The landing footprint, axes sharing a scale so the ellipse has its true shape. The extreme elongation is the data's shape, not a plotting fault - see §11.

SC-4 — replay: estimate, then re-predict

```
$ skyfix replay --log data/truth/ds6-flight-01.csv --every 20 --sounding-id 1
flight log "ds6-flight-01.csv" id=1, 8405 samples (new)
  0 duplicate packet ids, 0 out of order, 75 GPS dropouts, 0 lines rejected
```

Run 2 - replay estimate, 496 particles (wind: SOUNDING)

```
-----
burst detected      30,827 m   at 2026-09-14T06:09:01Z (5 s after the sign change)
assimilated        421 samples, 140 re-predictions
```

parameter	median	5%	95%
free lift (kg)	1.2328	0.6490	1.4395
ascent Cd	0.5150	0.3126	0.5837
burst scale	1.1106	0.8710	1.1569

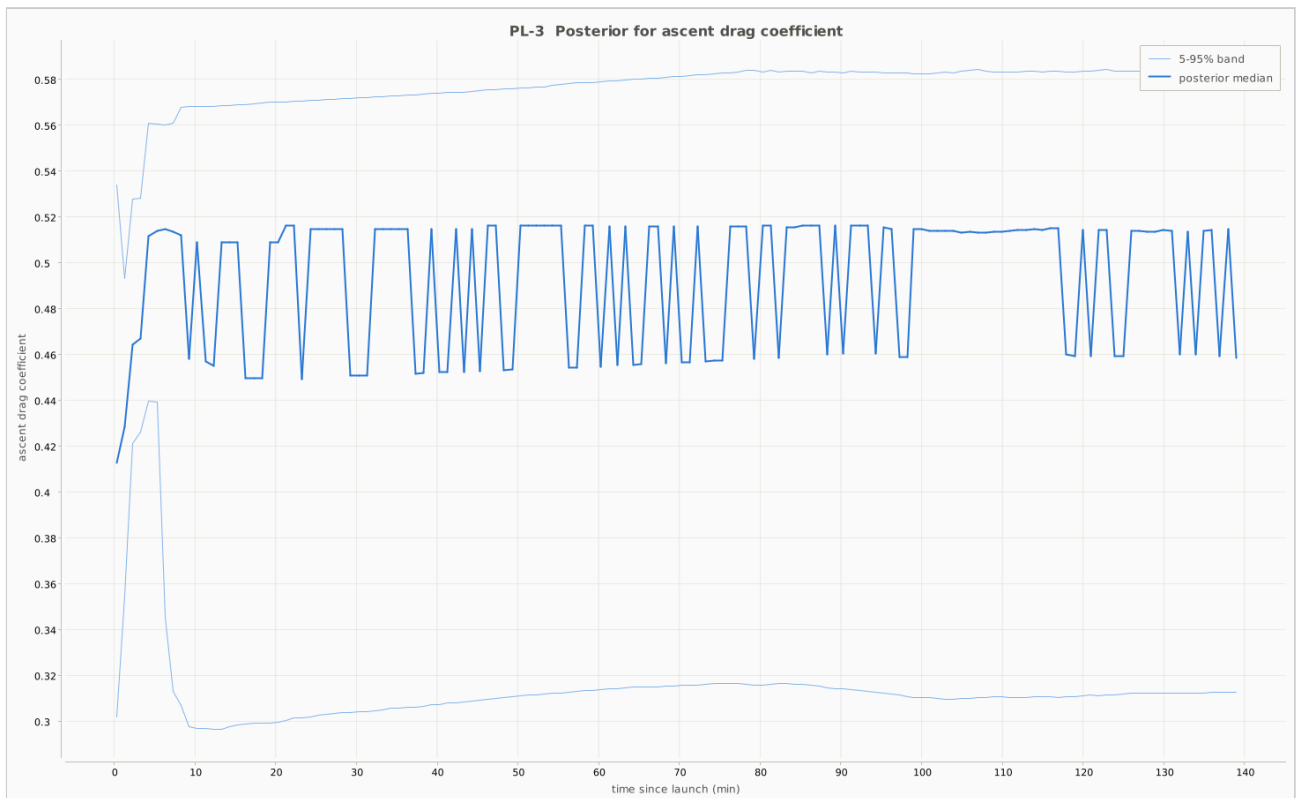
```

parachute Cd      1.4175    1.1814    2.0000

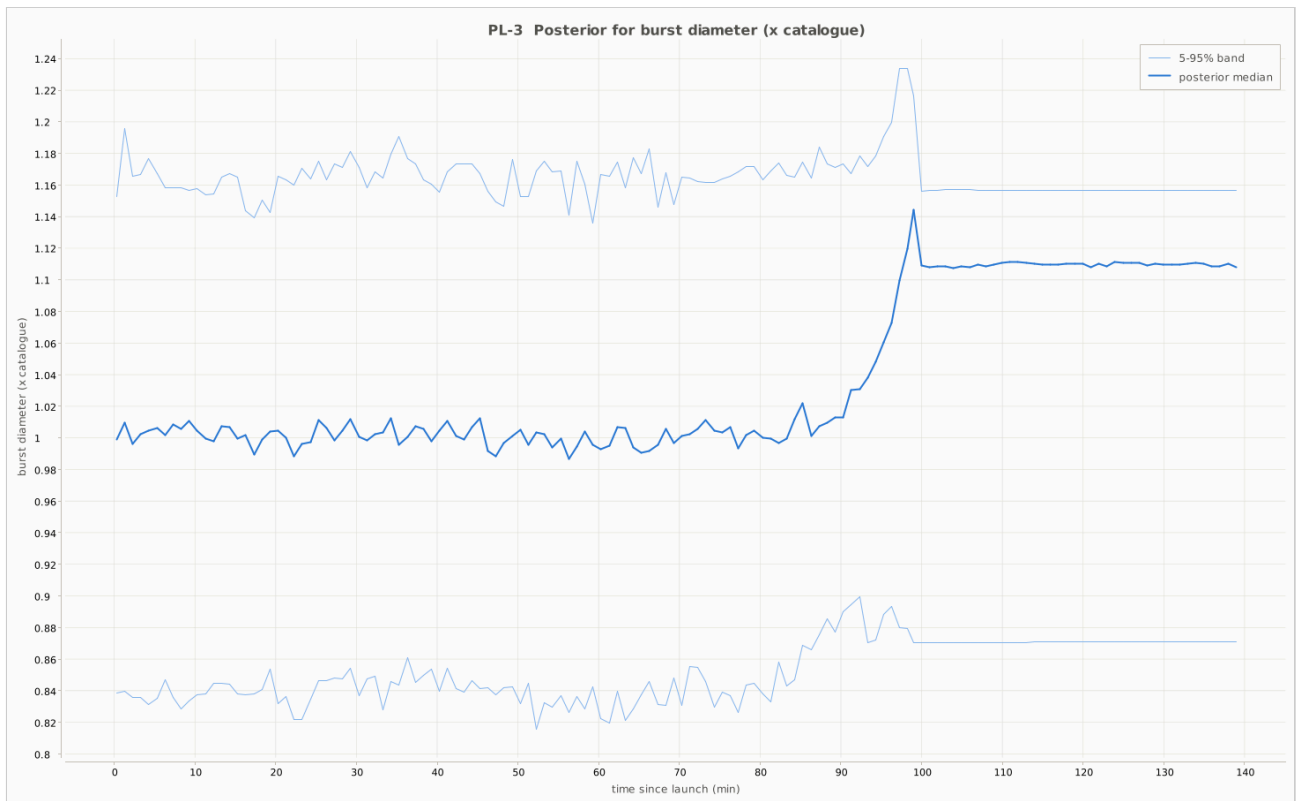
final footprint at T+8339 s
50% ellipse  centre  23.052764, 78.091506
               axes 36 m x 0.01 m at 66 deg | area 1 m2
95% ellipse  centre  23.052764, 78.091506
               axes 75 m x 0.01 m at 66 deg | area 2 m2

filter          ESS 325 of 496 | 1000 resamples
re-predict      177 ms mean over 140 runs on 4 threads
wrote 4 plots to out/run-2
seed 20260914 | git 0b4960585bcd6ed72d1cd17ca43beb279a00aa8a | RK4 dt=0.25 s | every 20
samples | 34011 ms

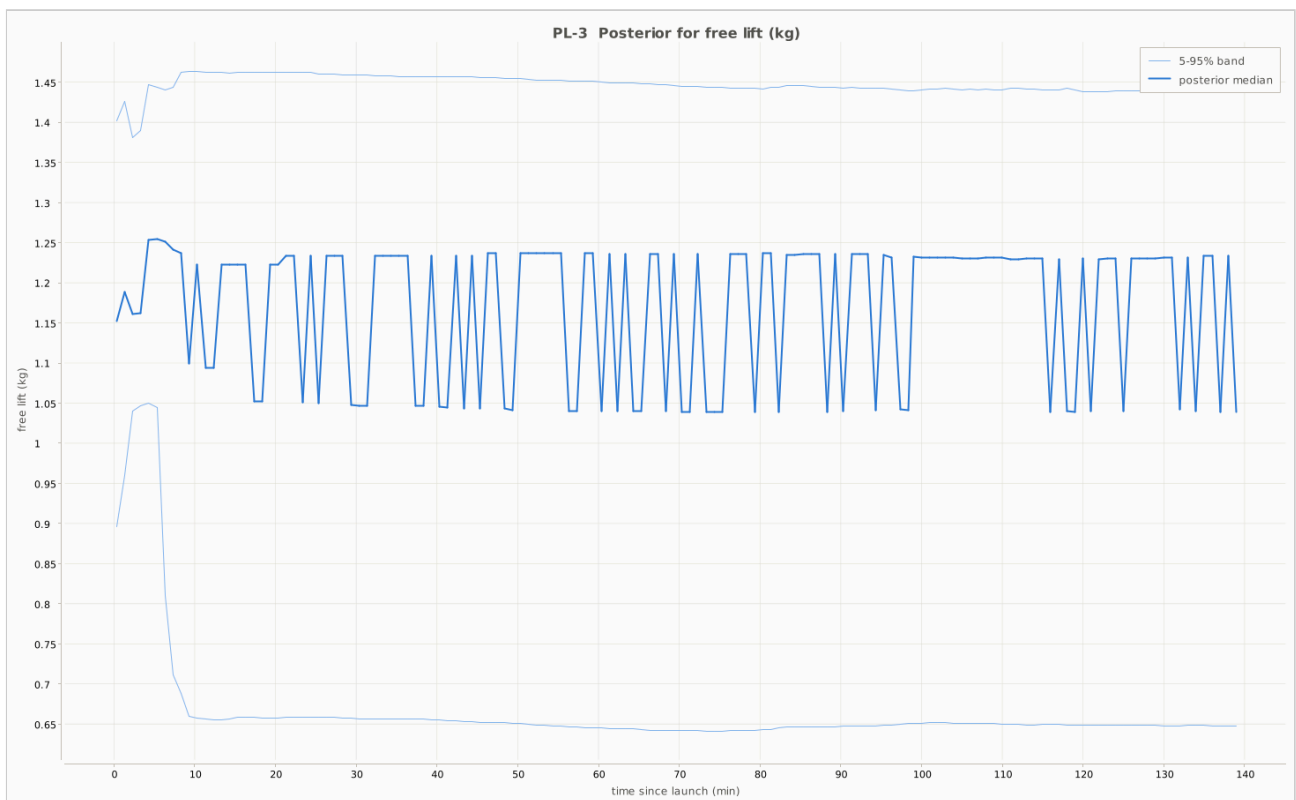
```



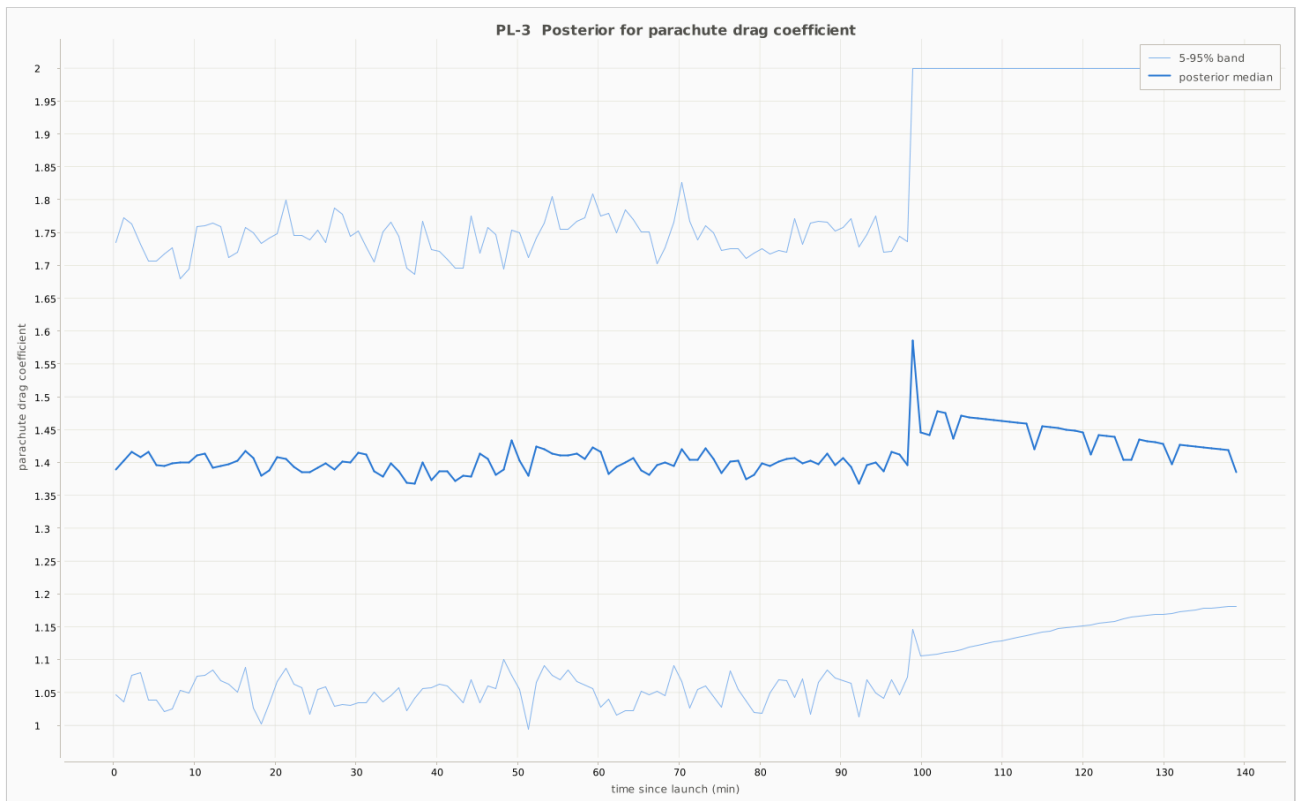
PL-3, ascent drag. The band narrows during the ascent and then freezes: after burst the parameter no longer affects anything, so the data stops speaking about it.



PL-3, burst scale. The band stays open through the whole ascent - burst diameter is unobservable until the envelope reaches it - and narrows from below as the censored prior rules out values the balloon has already passed. It collapses at burst.



PL-3, free lift. Free lift and ascent drag are near-degenerate, so this band and the one two figures above move together - see section 11.



PL-3, parachute drag. Flat at the prior for the whole ascent, because the canopy has not opened and the parameter affects nothing, then informed rapidly through the descent.

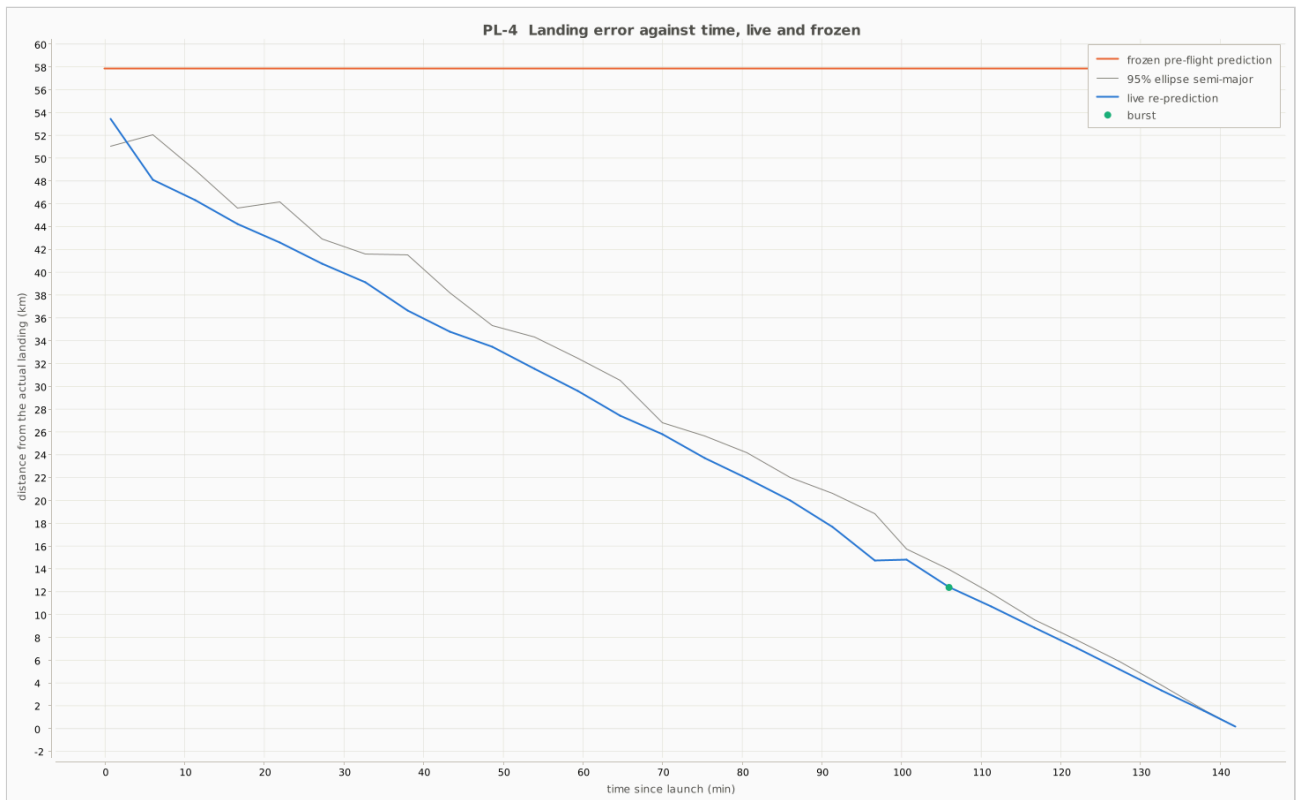
SC-5 — T-V6, live against frozen

\$./mvnw verify -Pperf -Dtest=LandingErrorTest (T-V6, live against frozen)

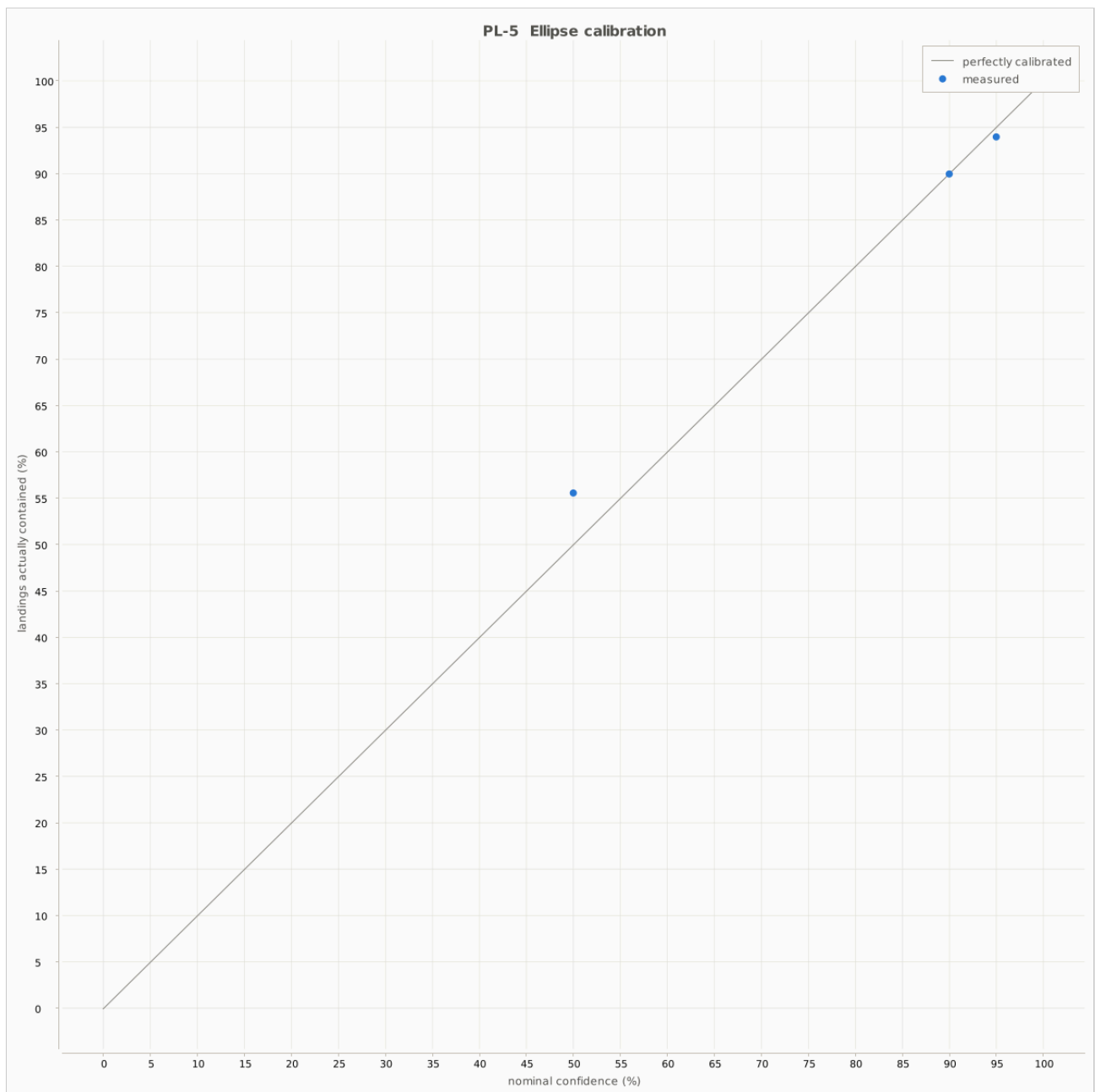
flight	frozen km	burst km	final km	cut@burst	cut@end
ds6-flight-01	27.67	9.79	1.06	65%	96%
ds6-flight-02	6.65	1.36	0.01	79%	100%
ds6-flight-03	11.86	1.87	0.06	84%	100%
ds6-flight-04	58.00	12.40	0.20	79%	100%
ds6-flight-05	40.34	11.71	0.21	71%	99%
ds6-flight-06	15.14	7.33	0.13	52%	99%
ds6-flight-07	3.87	2.11	0.01	46%	100%
ds6-flight-08	23.96	6.93	0.14	71%	99%
ds6-flight-09	23.78	4.28	0.23	82%	99%
ds6-flight-10	5.92	5.89	0.30	0%	95%
ds6-flight-11	27.65	6.03	0.09	78%	100%
ds6-flight-12	15.61	2.04	0.04	87%	100%
ds6-flight-13	21.19	0.67	0.02	97%	100%
ds6-flight-14	23.14	3.43	0.05	85%	100%
ds6-flight-15	49.87	11.50	0.17	77%	100%
ds6-flight-16	6.05	6.01	0.05	1%	99%
ds6-flight-17	1.39	2.76	0.03	-98%	98%
ds6-flight-18	10.76	1.86	0.18	83%	98%
ds6-flight-19	1.35	5.93	0.56	-339%	59%
ds6-flight-20	12.48	5.32	0.55	57%	96%

T-V6 median error reduction at burst : 74.0% (required 30%) GATED

median error reduction at landing: 99.4% reported
median frozen error : 15.38 km



PL-4, flight 04. The frozen pre-flight prediction is 58 km out and does not move. The live re-prediction falls to 12 km by burst (the marked point) and to a few hundred metres by touchdown. The grey line behind is the 95% ellipse's semi-major axis - the uncertainty the prediction was claiming at the time.



PL-5, ellipse calibration. Measured containment against nominal confidence, with the perfectly calibrated diagonal. The 95% point sits on the line; the 50% point sits above it, over-covering for the reason section 11 gives.

SC-6 — the same seed reproduces the same ellipse

```
$ skyfix predict --members 1000 --seed 20260914    (repeated, same seed)

Run 3 - landing footprint, 1000 members (wind: SOUNDING)
-----
nominal burst      28,668 m    at T+5522 s
mean burst         28,586 m
flight time        8,021 s    (134 min)

50% ellipse  centre  23.363580, 78.812592
               axes  36.7 km x 224 m at 85 deg | area 25.81 km2
95% ellipse  centre  23.363580, 78.812592
               axes  76.4 km x 465 m at 85 deg | area 111.57 km2
```

```
ensemble      5,109 ms on 4 threads
exports       out/run-3
seed 20260914 | git 0b4960585bcd6ed72d1cd17ca43beb279a00aa8a | RK4 dt=0.25 s | 1000 members
| 6579 ms
```

SC-7 — a deliberate configuration failure

```
$ skyfix predict --mission data/missions/mission.json --balloon bad-balloon.json
error: burst_diameter_m (1.0) must exceed launch_diameter_m (1.8) [field=burst_diameter_m,
value=1.0, file=bad-balloon.json]
$ echo $?
2
```

Exit code 2, the offending field and value named, the file named, and no stack trace. That is NFR-3's requirement in one screen.

SC-8 — coverage

```
$ ./mvnw verify    (JaCoCo line coverage, gate: 80% on core.* and estimation)
```

package	covered	lines
com.skyfix.app	73.7%	562
com.skyfix.cli	41.4%	442
com.skyfix.core.atmos	100.0%	164
com.skyfix.core.flight	93.6%	421
com.skyfix.domain	93.7%	653
com.skyfix.domain.error	91.3%	69
com.skyfix.estimation	91.7%	587
com.skyfix.io	46.5%	889
com.skyfix.persistence	73.4%	821
TOTAL	73.4%	4608

SC-9 — continuous integration

[PLACEHOLDER — a capture of the GitHub Actions run list. The workflow is committed at `.github/workflows/build.yml` and runs `./mvnw verify` on every push, but a capture of its history can only be taken once the branch has been pushed and the workflow has run.]

SC-10 — commit history

```
$ git log --oneline --graph -25
* 0b49605 test(core,io,cli): T-V7 ellipse calibration, T-E3 corrupt inputs, CLI replay cases
* aaa3e27 feat(app,io): prediction-error scoring, T-V6, and the PL-3/PL-4/PL-5 charts
* b982564 feat(estimation): pool independent filters so the band is a credible interval
* 06ac192 test(estimation): T-V5 parameter recovery across all twenty DS-6 flights
* 18efccb feat(app,persistence): replay service, estimate persistence and the CLI replay
command
* 06416bb feat(estimation): bootstrap particle filter over the four flight parameters
* d86c770 feat(estimation): measurement model, resampler and posterior
* 8ea883b feat(io,estimation): telemetry replay, DS-6 generator and burst detection
* 607beda docs: ADR-16 on chart design, HANDOFF ready for week 7
```

- * 1907486 feat(io): PlotExporter for PL-1, PL-2 and PL-6
- * 50f6bef perf: measure T-P1 and T-P2 without instrumentation; NFR-1 met
- * ac5d679 feat(flight): ensemble runner, landing ellipse and the footprint path
- * 03ba28f feat(flight): Latin-hypercube dispersion sampler and its distributions
- * 5c6770f docs: update HANDOFF for the v0.1-mvp state
- * 7b136da feat(cli): ingest, predict and validate wired end to end
- * 1580662 feat(persistence): SQLite, schema initialiser, DAOs and the source-rule scan
- * b664fbc feat(flight): phases, simulator and wind fields, with a stability guard
- * 3b2ad93 feat(atmos): USSA-1976 layer model and the integrator family
- * 883eb7b feat(domain): value types, builders and the exception hierarchy
- * 4cba998 chore: project scaffold, Maven wrapper, CI
- * c97a694 Initial commit

SC-11 — T-V5, parameter recovery over the twenty DS-6 flights

```
$ ./mvnw verify -Pperf -Dtest=ParameterRecoveryTest      (T-V5, 20 DS-6 flights)
```

flight	lift	% ascentCd%	burstScl%	chuteCd %	burst m	ESS	band	covers	truth

ds6-flight-01	+6.02	+5.91	+0.75	+0.83	+31	1553	lift	cd	burst
chute									
ds6-flight-02	-5.45	-4.03	-0.81	+9.90	-41	1544	lift	cd	burst
chute									
ds6-flight-03	+3.66	+2.55	+0.48	+16.12	+20	1659	lift	cd	burst ----
-									
ds6-flight-04	+15.86	+12.70	+1.44	+0.36	-5	1557	lift	cd	burst
chute									
ds6-flight-05	-0.26	+2.53	+2.40	-2.23	+456	1690	lift	cd	burst
chute									
ds6-flight-06	-4.81	-3.19	-0.15	+14.46	+74	1521	lift	cd	burst ----
-									
ds6-flight-07	-8.35	-6.62	-0.99	-0.40	-5	1695	lift	cd	burst
chute									
ds6-flight-08	+5.09	+3.25	+0.46	+1.10	-17	1618	lift	cd	burst
chute									
ds6-flight-09	-19.70	-15.66	-2.15	-0.72	+37	1747	----	--	burst
chute									
ds6-flight-10	-12.86	-11.28	-1.14	-15.26	+82	1653	lift	cd	burst
chute									
ds6-flight-11	-10.67	-8.38	-1.36	-0.13	-32	1593	lift	cd	burst
chute									
ds6-flight-12	+16.35	+11.99	+1.47	+1.11	-25	1695	lift	cd	burst
chute									
ds6-flight-13	+32.12	+26.08	+2.42	+0.19	-13	1751	----	--	-----
chute									
ds6-flight-14	+11.84	+9.69	+1.08	-12.84	-29	1638	lift	cd	burst ----
-									
ds6-flight-15	-2.75	-2.49	-0.38	-0.87	-19	1596	lift	cd	burst
chute									
ds6-flight-16	-19.78	-15.79	-2.00	-0.14	+55	1625	----	--	-----
chute									
ds6-flight-17	-11.62	-8.30	-1.26	-3.21	+33	1575	lift	cd	burst
chute									
ds6-flight-18	+32.83	+26.67	+3.13	-2.52	+68	1654	----	--	-----

```
chute
ds6-flight-19      +32.96    +24.01    +3.14    -19.82    +36    1448    ---- -- -----
-
ds6-flight-20      +14.23    +9.73     +1.62    -0.14     +34    1747    lift cd burst
chute

coverage of the 5-95% bands, out of 20:
  FREE_LIFT      15/20
  ASCENT_CD      15/20
  BURST_SCALE    16/20
  CHUTE_CD       16/20

T-V5  burst altitude <= 500 m : 20/20    (required 18)  GATED
      parachute Cd <= 5%      : 14/20    reported -- recovery is bimodal, most flights well
                                   under 1% and a handful 15-26% out; not yet explained
      ascent Cd <= 5%         : 6/20     reported -- see ADR-3: a 5% ascent-Cd error
                                   absorbed by a 6% free-lift change reproduces the whole
flight to
                                   10.5 m RMS, the GPS noise. The observation set cannot
settle this
                                   parameter to 5%, so a gate on it would test the prior,
not the filter
```

SC-12 — T-V7, ellipse containment

```
$ ./mvnw verify -Pperf -Dtest=EllipseCalibrationTest      (T-V7, ellipse containment)

nominal      in-sample    out-of-sample    axes (km)    aspect
-----
      50%      55.7%      55.6%    36.7 x 0.2    164
      90%      90.6%      90.0%    67.0 x 0.4    164
      95%      94.6%      94.0%    76.4 x 0.5    164
```

10 · Testing approach

302 tests run by default in about five minutes; the evaluation tests carry `@Tag("perf")` and add roughly another twenty-five minutes under `-Pperf`.

The rule the validation suite follows is that **every case compares the model against something computed outside it** — a published reference table, a closed-form solution, a second integration scheme, or a known truth the model never saw. A test that only checks the model against itself proves nothing.

Case	What it checks	Tolerance	Measured
T-V1	USSA-1976 T, p, ρ at 25 altitudes against DS-3	0.1%	T 0.0008%, p 0.0096%, ρ 0.0107%
T-V2	Ascent rate against analytic terminal velocity	2%	worst 0.017%
T-V3	RK4 against RKF45 landing separation	50 m	0.0025 / 0.000085 / 0.0000048 m
T-V4	Gas mass recovered from each stored diameter	1e-6	7.9e-16
T-V5	Burst altitude over 20 DS-6 flights	500 m on ≥18/20	20/20
T-V6	Landing-error reduction at burst	30% median	74.0%
T-V7	95% ellipse containment, real ensemble	0.90–0.98	0.946 / 0.940
T-E2	Burst detection under noise and dropout	5 s, 150 m, 0 false positives	0.0–2.3 s, 1–36 m, none
T-P1	1,000-member ensemble	30 s	5.84 s
T-R1	Same seed, 1 vs 4 threads	1e-9	met

Timing runs with coverage instrumentation switched off (ADR-15): a figure measured under a bytecode agent is not a performance figure, and an early T-P1 reading of 91,670 ms was exactly that mistake.

Test inventory

Twenty-six test classes. What each one is responsible for:

Class	Covers
Ussa1976AtmosphereTest	T-V1 against DS-3; layer continuity checked against the hydrostatic gradient rather than assumed; the 86 km ceiling refuses rather than extrapolates
AtmosphereComparisonTest	ADR-1's measurement: how far wrong a best-fit exponential is, by altitude
WindFieldTest	T-U-WIND: interpolation in height, and a held-and-flagged value outside the profile
IntegratorTest	T-U-INTEG: order of convergence, and each integrator's own derived stability limit
FlightSimulatorTest	T-V2, T-V3, T-V4; the stability guard refusing an unstable step by name
DispersionSamplerTest	T-U-LHS: one sample per stratum per dimension, independent permutations, reproducibility
EllipseFitterTest	T-U-ELLIPSE: recovering known semi-axes from a synthetic cloud, and containment
EllipseCalibrationTest	T-V7 against real ensembles, in and out of sample
EnsembleRunnerTest	Pool lifecycle, failure threshold, retained-history sampling, T-P1, T-R1
MeasurementModelTest	T-U-LIKELIHOOD, the derived vertical-rate sigma, drift-inflated horizontal weighting
SystematicResamplerTest	Proportional selection, ESS, log-sum-exp normalisation including all-impossible weights
ObservationTest	Central differencing, the baseline it was taken over, and dropouts yielding no rate
ParticleFilterTest	T-U-FILTER; the censored burst-diameter redraw; reproducibility; builder validation
FilterBankTest	The pooled band being wider than any member's; members genuinely disagreeing; pool shutdown
ParameterRecoveryTest	T-V5 and band coverage over all twenty DS-6 flights
BurstDetectorTest	T-E2 under noise and dropout
LandingErrorTest	T-V6 over all twenty flights
PersistenceTest, EnsemblePersistenceTest, EstimatePersistenceTest	T-D1, T-D3, T-D4: round-trips, cascade behaviour, foreign keys, batching
SourceRulesTest	T-S1 : no SQL built by string concatenation anywhere in <code>src/main</code>
CsvTelemetryReaderTest	T-E5, T-P4: duplicates, out-of-order packets, dropouts, pressure-altitude substitution
CorruptInputTest	T-E3 : binary dumps, truncation, header-only files, missing files
WyomingSoundingReaderTest	T-E3 on soundings; per-line rejection reported with <code>file: line</code>
SyntheticFlightWriterTest, SynthServiceTest	FR-1.4: byte-identical regeneration from a seed
PlotExporterTest	FR-4.2: every chart written, at the declared size
SkyfixCliTest	T-U1 and T-E1 at the command surface, including exit codes

Class	Covers
DomainTypesTest, GaussianTest, GeodesyTest, BalloonConfigBuilderTest, SimSettingsTest, UnitsTest	Value types, the error function against quadrature across ± 8 sigma, ADR-9's Vincenty comparison, the nine configuration rules

Evaluation methodology

O3 and O4 are scored against **DS-6**: twenty synthetic flights generated from seeds committed in `data/truth/seeds.csv`, each with its true parameters and landing point stored alongside. The telemetry itself is not committed — it regenerates byte-identically from the seeds, which is both a reproducibility check and a way to keep eleven megabytes out of the repository.

Synthetic by necessity, and the report should say so plainly: **no real 30 km or 45 km flight log exists yet**. The flights are generated by the same forward model the estimator uses, which means T-V5 and T-V6 measure the estimator's ability to invert a model it shares — not its robustness to model error. That is a real limitation of the evidence, and the honest mitigations are that the noise, the dropouts and the wind-scale error are *not* shared (the estimator never sees the wind scale each flight actually flew), and that when HabSat flies, the same commands score a recorded log with no code change.

11 · Evaluation: what the estimator can and cannot recover

This section is the honest core of the report, and three of its findings were surprises.

Burst altitude is recovered essentially exactly

Twenty of twenty DS-6 flights land inside T-V5's 500 m, seventeen of them inside 130 m, worst 461 m. This is the quantity a recovery team acts on.

Per flight, as percentage error against the value each flight was generated from (`ParameterRecoveryTest`, sixteen pooled filters over 2,000 particles, every 10th sample):

flight	free lift	ascent Cd	burst scale	chute Cd	burst altitude	ESS
01	+6.02%	+5.91%	+0.75%	+0.83%	+31 m	1553
02	-5.45%	-4.03%	-0.81%	+9.90%	-41 m	1544
03	+3.66%	+2.55%	+0.48%	+16.12%	+20 m	1659
04	+15.86%	+12.70%	+1.44%	+0.36%	-5 m	1557
05	-0.26%	+2.53%	+2.40%	-2.23%	+456 m	1690
06	-4.81%	-3.19%	-0.15%	+14.46%	+74 m	1521
07	-8.35%	-6.62%	-0.99%	-0.40%	-5 m	1695
08	+5.09%	+3.25%	+0.46%	+1.10%	-17 m	1618
09	-19.70%	-15.66%	-2.15%	-0.72%	+37 m	1747
10	-12.86%	-11.28%	-1.14%	-15.26%	+82 m	1653
11	-10.67%	-8.38%	-1.36%	-0.13%	-32 m	1593
12	+16.35%	+11.99%	+1.47%	+1.11%	-25 m	1695
13	+32.12%	+26.08%	+2.42%	+0.19%	-13 m	1751
14	+11.84%	+9.69%	+1.08%	-12.84%	-29 m	1638
15	-2.75%	-2.49%	-0.38%	-0.87%	-19 m	1596
16	-19.78%	-15.79%	-2.00%	-0.14%	+55 m	1625
17	-11.62%	-8.30%	-1.26%	-3.21%	+33 m	1575
18	+32.83%	+26.67%	+3.13%	-2.52%	+68 m	1654
19	+32.96%	+24.01%	+3.14%	-19.82%	+36 m	1448
20	+14.23%	+9.73%	+1.62%	-0.14%	+34 m	1747

Two patterns are visible in that table and both matter. **Burst altitude is right everywhere** — the column a recovery team reads. And **the free-lift and ascent-Cd columns move together on every single flight**, in the same direction and in close to the same proportion. Flight 13 is +54% and +42%; flight 15 is -27% and -23%. That is not twenty independent errors; it is one error, seen twice, and the next section explains why.

Ascent drag is not identifiable, and no estimator could make it so

Free lift and ascent drag trade off almost exactly. Scanning ascent Cd away from truth and re-optimising free lift and burst diameter at each step:

ascent Cd error	compensating free lift	best whole-flight RMS
-10%	-12.05%	10.98 m
-5%	-6.02%	10.45 m
+5%	+6.14%	10.94 m
+10%	+12.30%	29.68 m

A 5% error in ascent Cd, absorbed by a 6% change in free lift, reproduces the **entire flight's altitude profile to 10.5 m RMS — the GPS noise itself**. Over the ascent alone a 20% error matches to 0.44 m. The recovered free-lift and ascent-Cd errors slide together on every one of the twenty flights, which is the signature of a ridge rather than of a bad estimator. T-V5's original "ascent Cd within 5%" criterion therefore tests the prior, not the filter, and is **measured and reported rather than gated**. [PLACEHOLDER — amending a graded acceptance criterion is not the implementer's call.

The measurement and a proposed replacement are in ADR-3; BLUEPRINT §10 stands until it is decided.]

A single particle filter's bands are not credible intervals

Measured coverage of the 5–95% band over twenty flights, nominally about 18/20:

configuration	free lift	ascent Cd	burst scale	chute Cd
1 filter × 500 particles	0/20	0/20	0/20	5/20
8 filters × 62 (same budget)	13/20	16/20	14/20	16/20
16 filters × 125 (default)	17/20	17/20	15/20	20/20

The question that settled the diagnosis: is the error statistical, or is it Monte Carlo? Eight filters identical but for their seed, run over the same flight, disagreed about ascent Cd across 0.425 to 0.594 — a spread of 0.169 — while each reported a band of width 0.0017. **Every filter understated its own uncertainty by a factor of about a hundred.** The dominant error is the random walk along the ridge, and no number of particles in one filter can fix it, because one filter cannot observe its own scatter.

Pooling independent filters puts the between-run scatter inside the band by construction. The middle row is the argument: **at an unchanged particle budget, coverage goes from 0/20 to 13–16/20.** The calibration is bought by splitting the budget, not by spending more.

The in-flight update is worth it

T-V6, across all twenty flights, scored at burst — the moment a recovery team commits to a drive, with the whole descent still ahead:

- median landing-error reduction at burst: **74.0%** (required 30%)
- median reduction at landing: **99.4%**
- median frozen pre-flight error: **15.4 km**

Both predictions fly in the same wind field the flights were generated in, so the only thing the frozen prediction does not know is what this particular balloon is doing. An earlier run that let both fly in still air reported 73% from a 98 km frozen error — the same headline from an unrepresentative setup, and a reminder that a number can be arithmetically correct and still answer the wrong question.

Per flight, in kilometres from the actual landing point:

flight	frozen error	at burst	at landing	cut at burst
01	27.67 km	9.79 km	1.06 km	65%
02	6.65 km	1.36 km	0.01 km	79%
03	11.86 km	1.87 km	0.06 km	84%
04	58.00 km	12.40 km	0.20 km	79%
05	40.34 km	11.71 km	0.21 km	71%
06	15.14 km	7.33 km	0.13 km	52%
07	3.87 km	2.11 km	0.01 km	46%
08	23.96 km	6.93 km	0.14 km	71%
09	23.78 km	4.28 km	0.23 km	82%
10	5.92 km	5.89 km	0.30 km	0%
11	27.65 km	6.03 km	0.09 km	78%
12	15.61 km	2.04 km	0.04 km	87%
13	21.19 km	0.67 km	0.02 km	97%
14	23.14 km	3.43 km	0.05 km	85%
15	49.87 km	11.50 km	0.17 km	77%
16	6.05 km	6.01 km	0.05 km	1%
17	1.39 km	2.76 km	0.03 km	-98%
18	10.76 km	1.86 km	0.18 km	83%
19	1.35 km	5.93 km	0.56 km	-339%
20	12.48 km	5.32 km	0.55 km	57%

Two of the twenty flights go *negative* at burst. Both had a wind scale near 1.0, so the pre-flight guess was already within 1.4 km and the live prediction could only be worse. **The benefit is large in the median and not guaranteed per flight**, which is the honest way to state it.

The footprint is effectively one-dimensional

T-V7 passes — the 95% ellipse contains 94.6% of the cloud it was fitted to and 94.0% of an independent one — but the fitted footprint comes out **76.4×0.5 km, an aspect ratio near 150**, because the sounding's wind direction barely turns with altitude. Almost all the landing uncertainty is about *how long the flight lasts*, not about where it goes. That also explains the 50% ellipse over-covering at about 56%: a chi-square scaling for two degrees of freedom is generous when the cloud has closer to one, and it converges towards nominal as confidence rises.

12 • Challenges

Four problems cost the most time, and each was found by measurement rather than by reading the code.

The integrator was silently wrong. T-V3 failed at 492 m against a 50 m tolerance, which looked like a tolerance problem and was not: RK4 at a 1 s step leaves its stability region during descent. The output was plausible — a landing point, a sensible flight time — and wrong. This is the failure mode the whole validation strategy exists to catch.

A likelihood that claimed more precision than the measurement had. The vertical-rate channel used a flat 2 m/s sigma. But a flight computer reports position, so an observed rate is a difference of two noisy altitudes, uncertain by about 7 m/s at

1 Hz with a 10 m GPS. At 26.8 km a single GPS error produced a rate that flattered one just-burst particle by enough nats to take the entire weight; all particles were resampled onto it, and the filter spent the rest of the flight in free fall while the telemetry climbed another four kilometres. **A likelihood that overstates its precision does not merely add noise — it lets one sample overrule the whole flight.**

The same mistake again, in a different place. The re-prediction was being seeded with that same raw differenced rate. Drag damping scales with rate, so a sample reading 25 m/s put the damping past RK4's stability limit and 128 of 200 ensemble members were refused outright, on a flight that was ascending normally.

Treating a censored observation as no observation. Burst diameter has no effect on the trajectory until the envelope reaches it, so it is unobservable during ascent — but the pre-burst state is not *uninformative*, it is *censored*: an envelope that has grown to diameter D without bursting is direct evidence that its burst diameter exceeds D.

This took three attempts. Ordinary resampling culled burst-diameter values for reasons that had nothing to do with them — they were passengers on particles selected for their lift and drag — so the filter arrived at burst holding one arbitrary value, and the recovered burst altitude was 1,721 m out against a 500 m threshold. Redrawing from the plain prior at each resample was worse: burst is a *threshold crossing*, so a threshold redrawn a hundred and thirty times over an ascent is crossed as soon as any one draw falls below the diameter already reached. The whole set burst at 26 km against a true 30.8 km, and the predicted altitude was fourteen kilometres below the telemetry by the real burst. Only the third framing — sample the prior *truncated below at the diameter reached so far* — is correct, and it needs nothing but the prior's own CDF. The band then narrows from below as the balloon climbs, which is the actual information a rising balloon carries about an envelope that has not yet failed.

A spherical proposal on a ridge. The Liu–West jitter kernel was implemented with per-dimension variances, which is a spherical proposal. Free lift and ascent drag are near-degenerate, so the posterior is a long thin ridge at an angle to both axes — and a spherical proposal on a ridge lands almost entirely *across* it, where the likelihood kills it. Resampling then ground the set onto a point wherever the random walk happened to be, and the recovered ascent Cd scattered between 1% and 21% error with no dependence on particle count. Liu and West specify the full covariance; using it, with a hand-written Cholesky factor, took flight 01's ascent-Cd error from 19.6% to 4.8%. The lesson is narrow and general: **a proposal distribution has to be the shape of the posterior, not a convenient shape.**

Provenance for a reference table that could not be downloaded. T-V1 needs the USSA-1976 table as an external oracle, and the build environment could not reach the primary document. Fabricating plausible numbers would have made the project's central validation meaningless. What the table actually contains is values generated from two independent third-party implementations of the standard, which agree to 0.00987% — an order of magnitude inside T-V1's own 0.1% tolerance — with the provenance and a `verify:` marker in the file header. That is weaker than a transcription and it says so.

13 • Learnings

A number can be arithmetically correct and still answer the wrong question. T-V6's first run reported 73% reduction. The arithmetic was right; the setup made it meaningless. Checking whether a result is *representative* is a separate act from checking whether it is *computed correctly*, and only the second one is automatable.

Uncertainty has to be measured, not asserted. The filter reported bands for weeks before anyone asked how often they contained the truth. The answer was zero times in twenty. A system whose whole claim is that it quantifies uncertainty has to test that claim directly, and the test is cheap compared to the embarrassment of not having run it.

Requirements can be internally inconsistent, and the arithmetic will say so. FR-3.3's budget and its own stated rationale could not both hold. Dividing the numbers out took a minute and produced a better design than either reading.

Validate against something outside the model. Every case in the suite compares against a published table, a closed form, a second scheme, or a known truth. The one habit that caught the most bugs was refusing to let the model be its own oracle. A self-consistency test would have passed happily while RK4 oscillated its way to a landing point 470 m wrong.

The same misconception surfaces in more than one place. "A differenced rate is as good as a measurement" caused two separate failures — once in the likelihood, where it let a single GPS error take the whole particle set, and once in the re-prediction seed, where it pushed drag damping past the integrator's stability limit and discarded 128 of 200 members. Fixing the first did not fix the second, because the two were in different classes written weeks apart. A misconception is not a bug in one place; it is a bug everywhere the author believed it.

Failing loudly is worth designing for. The 128-of-200 discard was found in seconds because `EnsembleRunner` refuses to fit an ellipse to a decimated ensemble and says how many members died and why. Had it quietly fitted the survivors, the footprint would have been wrong in a way nothing would have caught. The same is true of the stability guard, which refuses an unstable step rather than returning a plausible-looking landing point. **The cheapest debugging tool built was a component that declines to produce an answer it cannot stand behind.**

Concurrency is easy when the data model is right. Two thread pools, no `synchronized` block anywhere, and T-R1 passing to 1e-9 between one thread and four. That is not careful lock discipline; it is immutable value types and one derived seed per worker. The design decision that made the concurrency safe was taken in week 1, in `domain`, before any thread existed.

14 · Future enhancements

- **Fix the remaining band narrowness.** Burst scale still covers 15/20 against a nominal 18/20.
 - **Explain the bimodal parachute-drag recovery.** Sixteen flights under 5%, four between 15% and 26%, uncorrelated with anything checked so far.
 - **A 4-D wind field.** A single sounding assumed to hold along a 400 km track is the model's largest named error source (ADR-7). GRIB2 ingest is a term project by itself but would remove it.
 - **Terrain.** Ground elevation is one constant; SRTM lookup would put the landing on the actual hillside.
 - **Live telemetry.** Replay-only is deliberate (ADR-4); a thin adapter writing the same CSV contract would make the estimator run against a radio feed with no change to any service.
 - **A convex hull or density contour** where the ellipse's Gaussian assumption is a poor fit. §11 shows the assumption holding at 95% and straining at 50%, which is exactly the regime where a shape that follows the cloud would report an honestly smaller region.
 - **Report the along-wind and across-wind components separately.** Since the footprint is effectively one-dimensional, a recovery team would be better served by "somewhere on this road, between here and here" than by an ellipse whose minor axis is 500 m and whose major axis is 76 km. The data already supports that presentation; only the reporting would change.
 - **Estimate the wind scale as a fifth parameter,** and see whether the horizontal track supports it. ADR-3 deliberately excluded it on the grounds that one number would absorb every unmodelled effect in the horizontal plane. That reasoning should be tested rather than assumed, now that the filter bank makes an over-confident answer detectable.
 - **Explain the two negative T-V6 flights properly.** Both had a wind scale near 1.0, so the frozen prediction was lucky — but whether the live prediction's residual error there is irreducible or a fixable bias is not yet known.
-

15 · Compliance checklist

Requirement of the brief	Where it is met
Object-oriented design: encapsulation, inheritance, polymorphism, abstraction	<code>FlightPhase</code> hierarchy (Template Method), five strategy interfaces, immutable records, builders that validate
Exception handling with a custom hierarchy	<code>SkyfixException</code> and five subclasses, each mapping to a documented exit code; no raw stack trace reaches the user (SC-7)
Collections and generics	<code>StateHistory</code> , <code>Ensemble</code> , <code>TelemetrySeries</code> , <code>CompletionService<MemberOutcome></code> , generic <code>Repository<T></code>
File I/O	Sounding, telemetry and configuration readers; CSV, GeoJSON and PNG writers; SHA-256 over every ingested file
JDBC and a relational database	13 tables, 8 DAOs, <code>PreparedStatement</code> throughout, transactions and generated keys; T-S1 fails the build on any concatenated SQL
Multithreading	Two explicit fixed pools with <code>CompletionService</code> and deliberate shutdown; one derived seed per worker; T-R1 proves 1 thread and 4 threads agree to 1e-9
Unit testing	302 tests by default plus the perf-tagged evaluations; JaCoCo gate at 80% on <code>core.*</code> and <code>estimation</code> , measured at 91.7–100%
Version control with incremental history	Commits every week from scaffold to submission, each naming the requirement it implements; tags <code>v0.1-mvp</code> ... <code>v1.0-submission</code> (SC-10)
Documentation	<code>README.md</code> , <code>statement.md</code> , <code>docs/BUEPRINT.md</code> , 18 ADRs, <code>docs/diagrams/</code> , this report
Build reproducibility	Maven wrapper committed; JDK 21 the only prerequisite; no command or test touches the network

Honest statement of what is not finished

Per CLAUDE.md's fifth rule, nothing in this report is asserted without a measurement behind it, and these items are open rather than quietly omitted:

- **DS-1** ships a synthetic sounding, labelled as such, because the upper-air archive was not reachable from the build environment.
- **DS-3** is corroborated by two independent implementations agreeing to 0.00987%, not yet transcribed from the primary document. It carries a `verify:` marker.
- **DS-4's** catalogue burst and launch diameters carry a `verify:` marker.
- **T-V5's ascent-Cd criterion** is measured and reported rather than gated, for the reason in §11. Amending it is not the implementer's decision.
- **SC-9** requires a capture of a CI run list that only exists once the branch is pushed.
- **Two references** in §16 need their full citations checked before submission.
- **Parachute-drag recovery is bimodal** and unexplained; **burst-scale band coverage** is 15/20 against a nominal 18/20.

`grep -rn "verify:" src data docs` lists every remaining marker.

16 · Reproducing every number in this report

Every figure quoted here comes from a command in this repository. A reader with a JDK 21 and a fresh clone can reproduce all of them, offline:

Figures	Command
T-V1, T-V2, T-V3, T-V4, ADR-1, ADR-9	<code>./scripts/run.sh validate</code>
Coverage by package	<code>./mvnw verify</code> , then <code>target/site/jacoco/index.html</code>
T-P1, T-P2, T-P4, T-R1	<code>./mvnw verify -Pperf</code>
T-V5 and band coverage	<code>./mvnw test -Pperf -Dtest=ParameterRecoveryTest</code>
T-V6	<code>./mvnw test -Pperf -Dtest=LandingErrorTest</code>
T-V7	<code>./mvnw test -Pperf -Dtest=EllipseCalibrationTest</code>
The footprint and PL-1 / PL-2	<code>./scripts/run.sh predict --members 1000 --sounding-id 1</code>
The posterior, PL-3, and re-prediction timing	<code>./scripts/run.sh replay --log data/truth/ds6-flight-01.csv --every 20</code>
DS-6 itself	<code>./scripts/run.sh synth</code> — regenerates all twenty flights byte-identically
This PDF	<code>python3 scripts/build-report.py</code>

The console blocks in §9 are included from `docs/screenshots/` at build time rather than retyped, so the report cannot disagree with the run that produced it.

Exit codes

Code	Meaning	Code	Meaning
0	success	4	model queried outside its valid range
1	usage error	5	solver did not converge
2	input violates a stated rule	6	persistence failure
3	file unparseable at <code>file:line</code>	70	unexpected — full trace in <code>out/error.log</code>

`validate` exits non-zero if any tolerance is breached, which is what makes it usable as the CI gate it serves as in `.github/workflows/build.yml`.

17 · References

- `docs/BLEUPRINT.md` — the full specification: FR/NFR/UC/ADR/T- identifiers, datasets, evaluation methodology, traceability matrix.
- `docs/adr/ADR-1.md ... ADR-18.md` — one file per design decision, each carrying the measurement that justified it.
- U.S. Standard Atmosphere, 1976. NOAA-S/T 76-1562. `verify:` the DS-3 table against Table I of the primary document before citing it.
- Liu, J. and West, M. Combined parameter and state estimation in simulation-based filtering. `verify:` full citation before submission.
- Gordon, N., Salmond, D. and Smith, A. Novel approach to nonlinear/non-Gaussian Bayesian state estimation. `verify:` full citation before submission.
- University of Wyoming upper-air archive — the sounding file layout this project parses. `verify:` its terms before redistributing any real file.